# Chapter 1

# Introduction

The research problem addressed in this thesis is the design, fabrication, and characterisation of a class of electronic devices — organic memristive devices based on ion-containing polymer composites — whose dynamical behaviour can be exploited as a computational resource. Because these devices occupy an unusual position at the intersection of solid-state electronics, polymer physical chemistry, and computational neuroscience, no single disciplinary perspective is sufficient to establish the context in which the rest of the thesis operates. The present chapter is therefore arranged as a sequence of complementary surveys. The remainder of this chapter addresses, in turn, the structural limitations of contemporary computing architectures and the motivation to move beyond them (section 1.1); the biological analogy from which neuromorphic engineering has historically drawn inspiration; the physics and taxonomy of memristive devices as candidate electronic primitives for post-von-Neumann computing; and finally the specific class of organic polymer-composite devices studied in this thesis, together with the explicit scientific objectives that the remaining chapters address. The statement of thesis objectives in the closing section fixes the scope of the experimental work and defines the criteria by which the contribution of this thesis is to be evaluated.

## 1.1 The Crisis in Modern Computing

The objective of the present section is to set out, in terms accessible to a reader trained primarily in physical chemistry rather than in computer architecture, why the hardware of modern digital computers is widely regarded as approaching a structural impasse, and why that impasse motivates the exploration of fundamentally different electronic primitives. Two lines of argument are pursued in parallel. The first, developed in section 1.1.1, traces the limitations of the stored-program architecture that has served as the basis of essentially all general-purpose computing machines since the late 1940s, and documents their quantitative manifestation in the present era as the *memory wall*, the *energy wall*, and the end of classical transistor scaling. The second, developed in section 1.1.2, surveys two distinct responses to those limitations that have emerged in the device-physics community: the integration of computation and storage within a single physical substrate (in-memory and analogue computing), and the abandonment of clocked, synchronous arithmetic in favour of event-driven, temporally structured operation. This second response — and in particular the identification of fading memory as a computational resource rather than an engineering defect — supplies the conceptual frame within which the devices reported in this thesis are to be evaluated.

### 1.1.1 Von Neumann Architecture and Its Limitations

The dominant architectural model underlying modern digital computers was articulated by von Neumann in the "First Draft of a Report on the EDVAC" [1], circulated in 1945 and formalised over the subsequent decade. Its defining feature is the *stored-program* concept: instructions and data are represented in the same memory and drawn from it, one word at a time, by a central processing unit under the control of an instruction pointer. This unification of code and data was not an incidental engineering choice but the enabling abstraction of general-purpose computing: a single physical machine becomes capable of implementing any computable function by loading a different sequence of binary words into memory. The resulting architecture — comprising a central processing unit, a main memory holding both code and data, and a bus connecting them — is so deeply embedded in the design of contemporary digital systems that it is usually referred to simply as *the* architecture of computing, and its constituent elements (cores, caches, dynamic random-access memory, instruction fetch, register file) are treated as unquestioned primitives in the design of processors for server, desktop, mobile, and embedded applications. More than seventy years of cumulative engineering effort have been expended on optimising every stage of this pipeline while leaving its basic topology essentially untouched.

A direct consequence of the stored-program model is that the processor and the memory are *physically distinct* subsystems, separated by a communication channel whose bandwidth is a primary determinant of overall system performance. Every elementary computational step — the fetch of an instruction, the load of an operand, the write-back of a result — requires one or more traversals of this channel. In a modern general-purpose processor, the channel is a hierarchy of caches bridging between register files clocked at several gigahertz and main memory implemented in dynamic random-access memory (DRAM), which operates at effective bandwidths one or two orders of magnitude lower. Even when the entire working set of a computation fits into cache, the separation of compute and storage imposes an unavoidable cost: data must be moved, in the form of electrical signals along interconnect metal, between the transistors that perform arithmetic and the capacitors or latches that hold state. No arithmetic operation can be performed without first transporting operands, and no result can be preserved without transporting it back. This fundamental pattern is usually referred to in the architecture literature as the *processor–memory dichotomy.*

In its simplest form, the von Neumann architecture exposes only a single port between processor and memory, so that at any given clock cycle a single word can be read or written but not both simultaneously. The resulting choke point is the *von Neumann bottleneck*, a term introduced by Backus in the late 1970s and used throughout the subsequent literature to describe the limitation that even an arbitrarily fast processor can make no more progress than the memory interface permits [2]. Decades of architectural innovation — pipelined instruction fetch, multi-level cache hierarchies, speculative execution, prefetching, out-of-order issue, banked and multi-ported memory subsystems, and dedicated coprocessors ranging from vector units to graphics processors — can be read as increasingly elaborate attempts to mitigate this bottleneck without replacing the underlying topology. Each such innovation preserves the fundamental distinction between compute and storage while partially hiding it from the software layer by amortising data movement over larger blocks, deeper queues, or more parallel channels. None removes the underlying asymmetry, and each carries its own overhead in silicon area, power, and design complexity. The cumulative effect is that the bottleneck has been displaced and renamed rather than dissolved; it reappears, in each successive architectural generation, at whichever interface happens to be slowest and most energy-expensive at that point in time.

The quantitative severity of the processor–memory imbalance was diagnosed explicitly by Wulf and McKee [3] in 1995, in an analysis that has come to be known as the *memory wall*. They observed that processor performance, as measured by the product of clock frequency and instructions per cycle, had been improving at an annual rate of approximately 60%, whereas the access latency and bandwidth of commodity dynamic memory had improved by close to an order of magnitude less per year. A simple arithmetic extrapolation of the two curves, taken from any plausible starting year, leads within a decade or two to a regime in which the average cycle is dominated almost entirely by memory access: the processor spends far more time waiting for operands than it does computing with them. Empirical evidence accumulated over the following two decades has confirmed this projection in detail. Modern microprocessors typically allocate between 30% and 60% of their total silicon area, and an even larger fraction of their power budget, to cache memory, prefetch logic, memory controllers, and interconnect, precisely because those resources are required to keep the arithmetic units supplied with data. Even in heavily cache-optimised numerical codes, a non-negligible fraction of cycles is spent stalled on memory operations, and the situation is considerably worse for workloads whose access pattern is irregular or whose working set exceeds the on-chip cache. The memory wall is thus not a speculative prediction but a measurable property of present-day systems, and one that scales unfavourably with computation size.

The memory wall as originally formulated was a performance argument. A second, and in several respects more fundamental, argument concerns energy. In a quantitative survey presented as a plenary address at the 2014 International Solid-State Circuits Conference, Horowitz summarised the cost of elementary operations in a representative modern CMOS process [4]. A 32-bit integer addition consumed of the order of 0.1 pJ; a 32-bit floating-point multiplication, a few picojoules; a 32-bit register-file read, also a few picojoules. By contrast, fetching a 32-bit word from on-chip static random-access memory cost tens of picojoules, and fetching the same word from off-chip DRAM cost several hundred picojoules to nanojoules — that is, three to four orders of magnitude more energy than the arithmetic operation that the word is fetched to enable. The consequence is that, in any workload dominated by data movement, the total energy budget is set not by the cost of computing results but by the cost of shuffling operands. This asymmetry is the *energy wall*, and it is not a transient property of a particular process node. It reflects an underlying physical fact: moving charge across a wire of non-trivial length dissipates energy in the wire's parasitic capacitance, and the total wire length in a memory subsystem greatly exceeds that within an arithmetic unit. The implication is direct: any architecture in which computation and storage are spatially separated is fundamentally bounded, in both speed and energy, by the cost of moving bits between them, and that cost does not scale away with transistor miniaturisation.

For several decades, the architectural inefficiencies catalogued above were absorbed, rather than eliminated, by the relentless exponential improvement of the underlying silicon technology. Two empirical scaling laws describe the regime in which that absorption was possible. The first, associated with Moore [5], observed that the number of transistors that can be economically integrated on a single chip doubles at roughly constant intervals of twelve to twenty-four months. The second, due to Dennard and co-workers [6], provided the physical rationale for why that doubling should translate into proportionate performance gains: if every linear dimension of a MOSFET is scaled by a factor $1/k$ and the supply voltage is scaled by the same factor, then the transistor switches $k$ times faster, consumes $1/k^2$ the energy per switching event, and requires $1/k^2$ the area, so that the power density of a circuit at constant activity remains invariant. When the two laws held simultaneously, each silicon generation delivered both more transistors and faster, cooler ones, and the delivered performance of a given microarchitecture could be improved without structural change. The memory wall and energy wall were then, in practice, perceived as distant rather than binding limits, because each generation of processor was fast enough to make its own memory subsystem look slower than the previous generation's, without that contrast proving fatal for the workloads of the day.

Dennard scaling, however, is known to have broken down progressively from approximately 2005 onwards, when the gate-oxide thickness approached a regime in which quantum-mechanical leakage currents dominate sub-threshold behaviour and prevent further reduction of the supply voltage at the rate required by the scaling theory [7]. Transistor count has continued to grow, but the associated performance and energy-efficiency gains per transistor have slowed markedly, and the power density of a uniformly active die has begun to increase with each generation — the regime commonly described as *post-Dennard* or, with an allusion to the resulting necessity of selectively deactivating portions of the die to remain within a thermal budget, the *dark silicon* era. Under these conditions, the previously tolerated inefficiencies of the von Neumann architecture become first-order constraints. The hardware community can no longer rely on a new process node to mask the cost of data movement; that cost must instead be addressed architecturally. Simultaneously, and essentially independently, the workloads that dominate modern computing have shifted away from the single-thread, cache-friendly scientific and business codes for which the classical memory hierarchy was optimised, and towards machine-learning training and inference kernels that are defined by enormous repeated matrix multiplications over parameter tensors resident in memory [8]. These workloads intensify, rather than relax, the data-movement bottleneck, and they do so at a historical moment at which the underlying silicon can no longer be relied upon to compensate.

The combination of these three trends — the structural dominance of data movement over arithmetic, the corresponding dominance of memory-access energy over arithmetic energy, and the end of the Dennardian compensation mechanism — is what justifies describing modern computing as facing a *structural* rather than an *incremental* problem. It is increasingly difficult to see further progress as a matter of refining the von Neumann pipeline; the architecture itself, and in particular its physical separation of computation from storage, is now understood as the central obstacle. This recognition is not a minority view: it has been articulated repeatedly, including in the Turing Award lecture of Hennessy and Patterson [7], in which the authors argue explicitly that the combination of the end of Dennard scaling and the rise of machine-learning workloads represents an inflection point comparable to the transition from vacuum tubes to transistors, and that the next decade of computer architecture will be defined by domain-specific hardware that co-locates memory and computation. It is against this background that the search for new electronic primitives — devices whose operation does not presuppose the traditional distinction between storage and processing, and whose physics is naturally suited to supporting large, low-energy data-dependent operations — has moved from an exotic research programme to a mainstream concern.

### 1.1.2 In-Memory and Event-Driven Computing Paradigms

Two broad responses to the situation described in section 1.1.1 have emerged within the device-physics and circuits communities over the past decade. Both abandon, in their most consistent formulations, the strict separation of computation and storage that defines the classical von Neumann machine, but they do so along different axes and with different computational targets. The first response retains the clocked, arithmetic character of digital computing but dissolves the spatial separation between memory and processor, performing arithmetic operations directly within (or immediately adjacent to) the array elements that hold the operands; this family of approaches is known collectively as *in-memory*, *near-memory*, or *processing-in-memory* computing. The second response retains, or even deepens, the spatial separation between different functional elements but abandons the synchronous, arithmetic model of computation in favour of operation driven by temporally structured, typically sparse, event streams, and exploits the intrinsic dynamical behaviour of physical devices as part of the computation itself. This second family is known as *event-driven* or *temporal* computing.

The central conceptual step of in-memory computing is the replacement of the classical *fetch–compute–write-back* loop by an operation in which the computation is implemented *physically* within a memory array. The canonical example is the analogue matrix–vector multiplication performed on a resistive crossbar. In such an array, a two-dimensional grid of memory cells holds conductance values that encode the entries of a matrix; the input vector is presented as a pattern of voltages applied to the rows, and the output vector appears as the column currents dictated by Kirchhoff's laws, each column current being the sum of row voltages multiplied by the corresponding conductances. A single clock period, and an energy cost dominated by the steady-state dissipation in the array rather than by any per-operation data transfer, thus yields a result that on a conventional processor would require as many multiply–accumulate operations as the matrix has entries. Demonstrations of this principle in hardware based on metal-oxide resistive memories [9], phase-change materials [10, 11], and organic ionic devices [12–14] have shown that matrix–vector products of practical size can be performed at energy efficiencies far beyond what digital accelerators based on standard CMOS logic can achieve. The same mechanism can be generalised to support convolutional operations, attention mechanisms, and other primitives central to contemporary machine learning. In this framing, the memory array is no longer a passive store but an active computational substrate, and the requirements placed on its individual cells are correspondingly different from those placed on the cells of a classical memory.

The device-level requirements imposed by the in-memory-computing paradigm, in its standard form, nevertheless remain recognisably those of a memory. A single cell should store a quasi-static conductance value that encodes a trained weight; that value should be set once (or updated occasionally during training) and then read, potentially many millions of times, without drifting significantly. The cell should be reproducible across a large array, with device-to-device and cycle-to-cycle variability kept below a tolerance set by the numerical precision demanded by the application. Non-volatility over operational timescales of seconds to years is a design goal: a cell that loses its stored value within milliseconds is, under these criteria, defective. When organic memristive devices are evaluated against this yardstick, the comparison is frequently unflattering. Polymer-composite ionic devices tend to show substantial device-to-device variability, conductance drift during repeated reads, and retention times of seconds to hours rather than years. An evaluation that takes the in-memory, non-volatile weight-storage criterion as its point of reference is therefore likely to classify such devices as promising but ultimately flawed competitors to metal-oxide or phase-change memories. It is important to recognise, however, that this is the evaluation criterion of a *specific* computational paradigm, and that alternative paradigms exist within which the same physical properties — short retention, device-to-device heterogeneity, dynamical relaxation — are not liabilities but resources.

The second response to the limitations of the von Neumann model, often developed in parallel to the first, begins from the observation that not all useful computation is most naturally expressed as a sequence of arithmetic operations on static operands. Many tasks of interest — the processing of natural auditory and visual signals, the control of continuous motor systems, the classification of biomedical waveforms, the extraction of features from streaming sensor data — are defined on inputs that are themselves time-varying and that carry information in their temporal structure. For such tasks, a natural computational model is one in which the inputs are represented as streams of *events* in continuous time, each event consisting of a channel identity and a timestamp, and in which the state of the computing system evolves continuously in response to these events rather than stepping through discrete clock cycles. This model is realised at the algorithmic level by spiking neural networks, at the circuit level by event-driven asynchronous digital logic, and at the device level by physical substrates whose intrinsic relaxation dynamics are of the same order of magnitude as the temporal structure of the inputs they process. It is this last level on which the present thesis operates.

The loose inspiration for this second paradigm is, characteristically, biological. Mammalian nervous systems perform demanding sensorimotor and cognitive tasks at power budgets several orders of magnitude below those of the most efficient artificial counterparts, and it is difficult to avoid the conclusion that at least part of this efficiency is attributable to architectural choices that differ systematically from the von Neumann model. Biological neurons emit discrete events (action potentials) sparsely in time; communicate through synapses whose conductance is itself a dynamical variable rather than a static weight; maintain and modify their state through slow, graded electrochemical processes rather than through explicit write operations; and are embedded in a physical medium (the extracellular ionic environment) whose properties are part of the computation. Although a direct transcription of these mechanisms into silicon is neither feasible nor obviously desirable, several architectural lessons are widely acknowledged as transferable: representing information as sparse events rather than dense words [15], exploiting slow device physics as a computational primitive rather than engineering it away, and treating device-to-device heterogeneity as a source of useful functional diversity rather than as noise to be suppressed. These lessons define the conceptual basis of neuromorphic engineering as a discipline, and they constitute the second route out of the von Neumann crisis pursued in the device-physics literature.

A particularly clean theoretical articulation of the computational value of dynamical devices was provided by Boyd and Chua [16], who showed that any time-invariant nonlinear operator with *fading memory* on a suitable function space can be uniformly approximated by a finite Volterra series. The statement carries a direct engineering corollary: a physical system whose present state carries a smoothly decaying trace of its past inputs can, when combined with a trainable linear readout, implement a broad class of useful time-varying computations. This insight was later operationalised, independently, by Jaeger in the form of the *echo state network* [17, 18] and by Maass, Natschläger and Markram in the form of the *liquid state machine* [19], and subsequently unified under the name *reservoir computing* [20, 21]. In the reservoir-computing paradigm, an untrained dynamical system — the reservoir — is driven by the input stream, and only a linear readout layer on top of its state is adjusted by learning. When the reservoir is realised in physical hardware rather than as a simulated recurrent network, the paradigm is called *physical reservoir computing* [22], and its hardware instances have included optical, mechanical, and spintronic systems as well as electronic devices whose internal dynamics exhibit the necessary fading-memory property. Demonstrations of physical reservoir computing in volatile memristive [23–25] and in organic electrochemical [26] substrates are of direct relevance to the device class studied in this thesis, and will be revisited when the computational framework is developed in a later chapter.

The central implication of this second paradigm for the present thesis is a reframing of the criteria by which a candidate memristive device is to be judged. Under the in-memory, non-volatile-weight paradigm, a short retention time is a defect: the device loses information that the application requires it to keep. Under the physical-reservoir and event-driven-computing paradigm, by contrast, a finite retention time is the very property that makes the device computationally useful, because it defines the effective window over which past inputs influence the present output — that is, the fading-memory timescale. Similarly, device-to-device variability, which is penalised under the weight-storage paradigm, becomes a resource under the reservoir paradigm: a bank of devices with a spread of intrinsic timescales, coupled to a common linear readout, can jointly resolve temporal structure over a wider range of scales than any single device could [22]. The same physical property — for example, the ionic relaxation time of a polymer electrolyte — that makes a device a poor non-volatile memory can therefore make it an excellent temporal computing primitive, provided that the architecture in which it is embedded is one that exploits, rather than suppresses, its dynamics. The work reported in this thesis adopts this second viewpoint throughout: the devices studied are not candidate non-volatile weights, and they are not evaluated as such. They are candidate heterogeneous, fading-memory elements, and the criteria by which they succeed or fail are those appropriate to that class.

Two broad commitments therefore shape the remainder of this chapter and the experimental programme to which it serves as an introduction. The first is that the computational target is not non-volatile weight storage within an analogue crossbar but temporal computing within an event-driven or reservoir-style architecture; the second is that the evaluation criteria applied to the devices studied here are those appropriate to temporal computing, and that comparisons to fast, high-endurance, non-volatile memories are set aside as category errors rather than as fair contests. The scientific contribution documented in the later chapters is, accordingly, structured in three layers: a single organic memristive device that is instrumented deeply enough to establish that the physical mechanism invoked in its design is in fact at work (reported in **??**); a broader comparative corpus of polymer-electrolyte devices, studied across composition and cation identity on a restricted but internally consistent set of dynamical measurements, that quantifies how the resulting fading-memory behaviour can be tuned by chemistry; and a data-driven demonstration that those chemistry-tuned dynamics can be exploited, in simulation on the measured data, as primitives of temporal computation. The remaining sections of the present chapter build up the physical and biological context required for this programme, and close with an explicit statement of the objectives against which the thesis is to be assessed.

## 1.2 The Biological Synapse as the Engineering Template

The argument developed in section 1.1 left open a concrete question: if the separation of storage and computation is the structural obstacle faced by digital computing, and if the appropriate replacement is not a faster memory or a larger cache but a new class of physical primitive, what should such a primitive actually do? The most productive source of an answer, and the one with which the neuromorphic-engineering community has worked for more than three decades, is biology. Mammalian nervous systems do not merely achieve comparable outcomes to digital machines at lower power: they do so using elementary computing units whose internal organisation flatly refuses the compute–storage dichotomy. In a biological circuit, the same physical element that propagates a signal also stores the history that gates its propagation, and the same element that stores the history is itself modified by the signals that pass through it. The element in question is the synapse. The purpose of this section is to establish, in a form accessible to a reader whose primary training is in chemistry or materials science, the minimal set of biological facts about synaptic signalling and plasticity that motivates the device-level objectives of this thesis, and to derive from those facts a list of functional requirements that a candidate hardware synapse should satisfy. Critically, the derivation presented here does not privilege persistent, high-retention behaviour over transient, decaying behaviour: biological synapses exhibit both, the distinction between the two is itself computationally meaningful, and any engineering template extracted from biology must accordingly leave room for volatile, fading-memory primitives alongside quasi-stable, programmable ones.

### 1.2.1 Neural Signal Transmission: A Brief Overview

The elementary cell of the nervous system, the neuron, is a morphologically polarised cell specialised for the reception, integration, and transmission of electrical signals. Its three principal compartments can be summarised in functional terms [27]. The *dendrites* form a branched input arbor that receives signals from other cells; the *soma* (cell body) integrates those signals and, provided the integrated membrane potential crosses a threshold, initiates an outgoing signal at the axon hillock; the *axon* propagates that signal, often over distances of millimetres to metres, to its terminal branches; and the *axon terminals* (boutons) release a chemical messenger onto the dendrites or soma of downstream cells. A mature mammalian neuron typically receives inputs from $10^3$–$10^4$ other neurons through specialised contacts at its dendrites. Each such contact is a *synapse*. Neural information processing at the circuit level is therefore the joint result of signal propagation along axons and signal transformation at synapses, and the cell's computational role is defined almost entirely by the pattern, strength, and dynamics of its synaptic contacts rather than by any intrinsic property of the soma in isolation.

The signal that travels along the axon is the *action potential*: a stereotyped, approximately millisecond-long depolarisation of the membrane potential generated by the voltage-gated opening of $Na^+$ channels followed by $K^+$-mediated repolarisation, and regenerated at each internodal patch along the axon by the same mechanism. Action potentials are all-or-nothing events: their waveform is essentially independent of the strength of the input that elicited them, and information is therefore not carried in the amplitude of any individual spike but in the *temporal* properties of spike trains — their rate, their precise timing, and their pattern of coincidence across neurons. This feature has a direct engineering consequence: the physical substrate of neural computation is *event-driven* in exactly the sense introduced in section 1.1.2. Between spikes, a neuron is quiescent and consumes only a leakage current; during a spike, it briefly dissipates a small, approximately fixed amount of energy. Useful computation is carried out by the patterned arrival of such events at the circuit's many synapses, not by a clocked update of a shared state register.

At the synapse, transmission is predominantly chemical rather than electrical. When an action potential reaches an axon terminal, the depolarisation opens voltage-gated $Ca^{2+}$ channels; the resulting influx of $Ca^{2+}$ triggers the fusion of a small number of vesicles, each containing a quantised amount of a neurotransmitter molecule (for example glutamate, the dominant excitatory transmitter in the cortex), with the pre-synaptic membrane. The neurotransmitter diffuses across the narrow *synaptic cleft* ($\sim$20 nm) and binds specifically to receptor proteins on the post-synaptic membrane. Receptor binding causes, directly or via second-messenger cascades, the opening of ion channels in the post-synaptic membrane, and the consequent ionic current — the *excitatory* or *inhibitory post-synaptic current*, EPSC or IPSC — transiently shifts the post-synaptic membrane potential. The amplitude of this current, together with its kinetics, defines the *strength* of the synapse at that moment. The chain of events as a whole is probabilistic, quantal (the number of vesicles released per spike is a small integer subject to shot noise), and chemically rich: the same molecular machinery that propagates the signal can, through modulation of its many components, change the strength of subsequent transmissions.

The synapse is therefore the functional locus at which several distinct processes coincide. It is the point of *signal transmission* between neurons; the point of *signal modulation*, through the regulated amplitude and time course of the post-synaptic current; and the point of *memory*, through long-lived changes in that amplitude in response to the cell's own past activity. This co-localisation of transmission, modulation, and memory within a single physical element is not incidental: it is the reason the synapse is widely regarded as the fundamental computational unit of the nervous system, and it is the feature that makes the synapse, rather than the neuron, the most informative biological template for post-von-Neumann electronic devices.

### 1.2.2 Synaptic Plasticity: The Physical Basis of Learning

That experience leaves a physical trace in the nervous system, and that this trace is preferentially localised at synapses, is one of the oldest and most thoroughly supported hypotheses in modern neuroscience. Its conceptual formulation is due to Hebb [28], whose 1949 monograph argued that if a pre-synaptic cell repeatedly participates in firing a post-synaptic cell, the connection between the two should be strengthened. The modern précis of this idea — "cells that fire together, wire together" — captures its central structural claim: the relevant learning rule is *local* (it depends only on the activity of the two cells directly coupled by the synapse) and *correlation-sensitive* (it is driven by joint, rather than independent, activity). As a physiological hypothesis, Hebb's postulate remained without direct experimental test for more than two decades, but it has since come to organise essentially every subsequent account of synaptic learning, and the various forms of *synaptic plasticity* described below can all be read as mechanistic instantiations of Hebbian, anti-Hebbian, or Hebbian-like rules operating at different timescales.

The canonical demonstration that synaptic strength can be persistently altered by patterned activity was provided by Bliss and Lømo [29], who showed that a brief, high-frequency tetanic stimulation of the perforant path of the anaesthetised rabbit hippocampus produced an increase in the amplitude of the population post-synaptic potential that persisted for hours and, in subsequent studies, for days to weeks. This phenomenon, known as *long-term potentiation* (LTP), has become the paradigm of persistent, experience-dependent synaptic change. Its mechanistic basis in the hippocampus involves the NMDA-receptor-mediated coincidence detection of pre- and post-synaptic activity, a $Ca^{2+}$-dependent signalling cascade, and the regulated insertion of additional AMPA receptors into the post-synaptic membrane, which increases the amplitude of the EPSC elicited by a subsequent pre-synaptic spike. A complementary phenomenon, *long-term depression* (LTD), produces an equally persistent decrease in synaptic strength in response to low-frequency or anti-correlated activity, and is understood to involve the reverse trafficking of AMPA receptors and analogous signalling steps of opposite sign. Together, LTP and LTD provide a bidirectional, long-lasting modification of synaptic amplitude that is widely regarded as the physiological substrate of declarative and procedural learning.

Superimposed on these long-lasting changes, and arguably of greater importance for moment-to-moment information processing, is a family of rapid and reversible modulations known collectively as *short-term plasticity* [30]. Following a pre-synaptic action potential, the immediate strength of the synapse depends on the recent history of its own activity: *short-term facilitation* is a transient increase in EPSC amplitude in response to a second spike closely following the first, attributable to residual pre-synaptic $Ca^{2+}$ that has not yet been cleared from the terminal; *short-term depression* is a transient decrease attributable to vesicle depletion or receptor desensitisation after recent heavy release. Characteristic time constants for these effects range from tens of milliseconds to several seconds, with different synapses exhibiting different mixes of facilitation and depression. The functional significance of short-term plasticity is substantial: it implements temporal filtering at the level of the individual synapse, so that the same sequence of pre-synaptic spikes produces different post-synaptic responses depending on the inter-spike intervals within that sequence. Short-term plasticity turns the synapse into a nonlinear, history-dependent temporal filter, and it does so through entirely transient, self-resetting changes in state that never consolidate into a permanent modification of the connection.

A third and conceptually unifying form of plasticity is *spike-timing-dependent plasticity* (STDP), described in cortical pyramidal cells by Markram and co-workers [31] and subsequently characterised in hippocampal cultures in quantitative detail by Bi and Poo [32]. STDP sharpens Hebb's postulate into a timing-sensitive form: whether a given synapse is potentiated or depressed by a pairing of pre- and post-synaptic spikes depends on the sign and magnitude of the interval between them, typically on a window of a few tens of milliseconds. When the pre-synaptic spike reliably precedes the post-synaptic spike, the synapse is potentiated; when the order is reversed, it is depressed; when the interval is long in either direction, no lasting change is induced. The STDP window is therefore an implementation, at the level of a single synapse, of a directional correlation detector whose output is a long-lived change in synaptic efficacy. It is significant for the present thesis that the STDP rule is inherently temporal: it requires the synapse to retain a short-lived trace of its own recent pre-synaptic activity long enough to be compared to the post-synaptic response, and then to convert that comparison into a persistent modification of amplitude.

The picture of synaptic behaviour that emerges from these three lines of evidence is therefore not that of a single variable passively storing a "weight" but of a stateful element exhibiting a hierarchy of coexisting, physically distinct timescales. On the millisecond timescale, the synapse transmits individual spikes through a quantal release process. On the tens-to-thousands-of-milliseconds timescale, short-term facilitation and depression transiently amplify or attenuate that transmission in a way that depends on recent activity. On the timescale of minutes to hours and longer, LTP and LTD produce persistent, experience-dependent changes in baseline amplitude. All three layers coexist at essentially every chemical synapse in the mammalian central nervous system, and they are mechanistically supported by partially overlapping but distinct molecular machinery [33]. The implication for an engineering reader is important: memory in the nervous system is not a single phenomenon but a stratified set of phenomena, in which decaying, reversible, transient changes of state are integral computational elements rather than imperfect approximations to some idealised non-volatile store. Abbott and Regehr [33] argued on exactly these grounds that the synapse is better understood as a *computational* element than as a mere memory element: the dynamics of short-term plasticity, for example, allow a single synapse to act as a high-pass, low-pass, or band-pass temporal filter depending on the mix of facilitation and depression it exhibits, and the trained circuit as a whole exploits such filters in parallel. A hardware template that takes biology seriously must therefore encode both durable and decaying state changes within the same class of device, and must regard the presence of forgetting as a feature to be tuned rather than as a defect to be eliminated.

### 1.2.3 Why Hardware Emulation of the Synapse?

It is instructive, at this point, to consider why a hardware embodiment of synaptic function is needed at all, given that the mathematical abstractions of neural computation — weighted sums, nonlinear activations, gradient-based weight adjustment — can be and routinely are implemented in software on conventional processors. The answer follows immediately from the argument of section 1.1. A software neural network, whether it is an artificial feed-forward model or a spiking simulation, is executed as a sequence of arithmetic operations on values drawn from and written back to a memory hierarchy that is physically distinct from the arithmetic units. Every forward pass of such a network therefore pays the memory-wall and energy-wall costs catalogued above, and those costs scale with the number of synapses being simulated. The cortex of a small mammal contains of order $10^{12}$–$10^{14}$ synapses; the operating power of its nervous system is of order 10 W. No present or near-term von Neumann substrate is likely to close that efficiency gap, because the cost of fetching a weight from remote memory before multiplying it by an activation is fundamentally bounded by the interconnect physics of the system, not by further microarchitectural tuning. The conclusion of the previous section was that any convincing resolution must co-locate memory and computation in the same physical device. The conclusion of the present section is that the synapse is precisely such a device, and that it remains the most compact, lowest-power, and most functionally rich instance of co-localised analogue memory and computation known. An engineering programme that takes the synapse as its physical template, rather than as a merely mathematical abstraction, is therefore a natural response to the situation diagnosed in section 1.1.

The research programme that pursues this observation in circuit form is *neuromorphic engineering*, a field whose programmatic origin is usually attributed to Mead [34], and whose defining ambition is to build electronic circuits whose devices, topology, and dynamics physically instantiate the computational primitives of the nervous system rather than emulate them in software. Within this programme, the synapse has emerged as the key device-level target, and the search for a physical "artificial synapse" has driven a substantial body of work across device physics, materials science, and circuit design over the past two decades. The goals of that search, distilled from a large review literature [15, 35], can be stated relatively compactly. A hardware synapse should support an analogue, multi-state conductance that can be set reversibly by applied electrical stimuli, read non-destructively through a small probe current, and arrayed at high density so that many thousands of such elements can be addressed from a manageable number of input and output lines. Its response to applied stimuli should be nonlinear in a biologically meaningful sense — for instance, exhibiting a pulse-number or pulse-amplitude threshold below which no change is induced — and its dynamics should be compatible with the timescales of the signals it is meant to process, which span several orders of magnitude from milliseconds to seconds and beyond.

Among these requirements, the demand for *analogue*, rather than binary, multi-state conductance modulation deserves particular emphasis. A binary synapse encodes one bit of information per element and, by the standard argument of rate coding, requires prohibitively many elements to represent the graded synaptic weights of biological circuits. An analogue synapse, by contrast, can in principle encode several bits per element through the available range of conductance values, and can do so at a write energy that, for ionic-conductance devices of the class considered in this thesis, is several orders of magnitude below the energy cost of an equivalent digital multi-bit write. Reversibility is equally important: a synapse whose conductance can be potentiated but not reliably depressed is incapable of supporting the bidirectional Hebbian rules discussed in section 1.2.2, and is therefore useful only as a programmable weight in a feed-forward classifier, not as a substrate for on-chip learning. It is this combination of properties — analogue, reversible, multi-state, low-energy conductance modulation under electrical pulses — that has historically been taken as the first-order specification of a useful hardware synapse, and it is this specification that motivates the class of devices considered in the remaining chapters of this thesis.

At this point, however, a tension arises between a naïve reading of that specification and the lessons extracted from biology in section 1.2.2. Reviews of the artificial-synapse literature have often framed the target as a non-volatile, high-retention analogue memory whose stored conductance persists indefinitely once written, and have treated the retention time as a figure of merit to be maximised. This framing is defensible if the hardware synapse is taken to represent *only* the quasi-stable component of biological plasticity — that is, the component captured by LTP and LTD. It is, however, incomplete. Biological synapses are simultaneously quasi-stable stores of long-term potentiation and transient, reversible, decaying filters for short-term plasticity and spike-timing-sensitive integration; they coexist in the same physical element, and they contribute jointly to the computation performed by the circuit. A hardware template that captures only the quasi-stable component is therefore not, strictly speaking, a hardware synapse: it is a hardware analogue of *half* of a synapse. A complete engineering template should leave room for two complementary device roles: *persistent or quasi-persistent programmable weights* that stand in for the long-term component of plasticity and serve as slowly modified parameters of the network, and *dynamic fading-memory elements* that stand in for the short-term component and provide the temporally structured, history-dependent responses required for event-driven temporal computing.

The second of these two roles is the one of primary interest in this thesis, and is also the one most often under-represented in retention-centric framings of the artificial-synapse problem. A volatile device, in this role, is not a defective non-volatile memory: it is an intrinsically useful element whose physics supplies, without additional circuitry, the kind of leaky integration, temporal filtering, and exponential-decay-of-recent-history responses that short-term plasticity implements in biology. The continuity with the paradigm of section 1.1.2 is direct. A bank of such devices, with a spread of intrinsic time constants dictated by their physical composition, constitutes exactly the heterogeneous reservoir of fading-memory elements that physical reservoir computing requires [22–25]. In this context, forgetting is not a failure mode but a computational necessity: it defines the effective temporal receptive field of each device, it prevents the uncontrolled accumulation of history that would otherwise saturate any finite-range conductance, and it decorrelates the device's present state from the distant past in a way that keeps the reservoir responsive to recent inputs. A hardware synapse that cannot forget, in this sense, is strictly less useful for temporal processing than one whose physics includes a well-controlled decay timescale. The biological template, properly read, supports precisely this conclusion: persistent and transient state changes are both integral, and any responsible engineering transcription of synaptic function must admit device classes of both kinds.

These considerations narrow the search for a material platform. A candidate hardware synapse should support analogue, history-dependent, reversible modulation of conductance by electrical stimuli; it should operate at low energy per update; it should be compatible with dense, regular array integration; and — critically for the present thesis — it should admit device variants whose conductance state decays on biologically relevant timescales as well as variants whose state persists indefinitely. An additional, less technical desideratum is chemical tunability: because the timescales of short-term plasticity in biology are set by the kinetics of $Ca^{2+}$ buffering, vesicle recycling, and receptor desensitisation, and because these kinetics differ measurably between synapse types, any convincing hardware template should provide a chemical handle through which the corresponding electronic timescales can be engineered. No classical transistor-based memory structure meets this entire list of requirements simultaneously. The question that opens the next section is therefore natural: is there a physical device class that does?

## 1.3 The Memristor: Theory and History

The question that closed section 1.2.3 has a partial but important answer. There exists in circuit theory a concept, and in the experimental literature a diverse but coherent family of electronic devices associated with that concept, whose defining feature is precisely that the instantaneous current–voltage relation depends on an internal state imposed by past activity. The concept is that of the *memristor*; the family is that of *memristive devices*. The present section develops the theoretical content of the memristor concept in the minimum depth required for a thesis introduction, reviews the experimental realisation that established the concept as a physical rather than merely formal object, classifies the principal mechanistic families in which memristive behaviour is currently implemented, and closes with the set of parameters by which memristive devices are conventionally judged together with the reframing of those parameters demanded by the volatile, heterogeneous devices that are the subject of this thesis. A secondary and equally deliberate purpose of the section is to guard against a narrow reading of the memristor concept that identifies it with *non-volatile resistive memory*: as will be argued below, the formal definition of a memristive system accommodates both quasi-stable and relaxing internal states, and the field has progressively come to recognise that volatile memristors are not defective members of the family but an architecturally distinct subclass whose computational applications are different from, and not inferior to, those of their non-volatile counterparts.

### 1.3.1 Theoretical Foundation

Classical linear circuit theory describes a two-terminal passive element through the relation it imposes between two of the four fundamental lumped variables: the voltage $v$, the current $i$, the electric charge $q = \int i\,\mathrm{d}t$, and the flux linkage $\varphi = \int v\,\mathrm{d}t$. Three of the six possible pairings among these four variables are occupied by the familiar elements of elementary electromagnetism. The resistor $R$ is defined by the direct $v$–$i$ relation $v = R\,i$; the capacitor $C$ by the $q$–$v$ relation $q = C\,v$; and the inductor $L$ by the $\varphi$–$i$ relation $\varphi = L\,i$. Two further pairings, between $q$ and $i$ and between $\varphi$ and $v$, are trivial consequences of the definitions of charge and flux linkage as time integrals. The remaining non-trivial pairing, that between charge $q$ and flux linkage $\varphi$, has no classical element associated with it.

The gap was filled by Chua [36] in 1971, who observed that completeness of the passive-element taxonomy required a fourth fundamental two-terminal element relating charge to flux linkage. He named it the *memristor*, a contraction of "memory resistor", and defined it by the constitutive relation $\varphi = \hat{\varphi}(q)$ or, in an equivalent form used more widely in the subsequent device literature, by $v = M(q)\, i$, where the *memristance* $M(q) = \mathrm{d}\hat{\varphi}/\mathrm{d}q$ has the units of a resistance but depends functionally on the charge that has flowed through the device. The operational content of this equation is that a memristor is a resistance whose present value is determined by the history of the current that has passed through it. If $M$ is a constant, the device reduces to an ordinary resistor; if $M$ depends non-trivially on $q$, the device retains a memory of past activity in its present electrical response. The universal fingerprint of such behaviour in the $i$–$v$ plane is a *pinched hysteresis loop* passing through the origin under periodic excitation, reflecting the fact that zero current requires zero voltage instantaneously while the device nonetheless carries a history that biases its response to subsequent stimuli. This hysteretic fingerprint is the elementary experimental criterion by which memristive behaviour is diagnosed throughout the device literature.

The ideal memristor defined by $\varphi = \hat{\varphi}(q)$ is, however, a restrictive mathematical object, and very few physical devices satisfy it exactly. Chua and Kang [37] therefore generalised the concept in 1976 to that of a *memristive system*, specified by the pair of equations

$$\begin{aligned} \dot{\mathbf{x}} &= f(\mathbf{x}, u, t), \\ y &= g(\mathbf{x}, u, t)\, u, \end{aligned} \tag{1.1}$$

in which $u$ and $y$ are the electrical input and output (typically voltage and current, or their dual), $\mathbf{x}$ is an internal state vector that captures the memory of the device, and $f$ and $g$ are generally nonlinear functions. The formal implication of eq. (1.1) is central to the argument of this thesis. The state $\mathbf{x}$ is free to evolve under its own dynamics, and in particular the state equation $\dot{\mathbf{x}} = f(\mathbf{x}, u, t)$ may include relaxation terms that drive the state towards a rest value once the driving input is removed. A device whose state relaxes is, by this definition, a memristive system, even though it is not an ideal memristor in the strict sense of [36]. The broader memristive-systems framework therefore explicitly accommodates volatile devices — devices whose memory of past activity decays in time — and treats them as first-class members of the family rather than as defective approximations to an idealised non-volatile store. It is within this broader framework, and specifically in the relaxing-state sector of the framework, that the devices reported in the remaining chapters of this thesis are to be understood.

### 1.3.2 The First Physical Realisation: $TiO_2$ Memristor

For more than three decades after Chua's original proposal, the memristor existed principally as a theoretical construct. Thin-film metal-oxide resistive switches exhibiting hysteretic current–voltage characteristics had been described in the device-physics literature since the late 1960s, and in retrospect many of these systems can be classified as memristive. None, however, had been interpreted explicitly within Chua's formalism, and the field lacked the benefit of a single canonical experimental realisation that would tie the formal concept to a concrete thin-film device. The identification was made by Strukov, Snider, Stewart and Williams [38] in 2008, working at HP Laboratories. Their device consisted of a thin film of titanium dioxide ($TiO_2$) sandwiched between two platinum electrodes, in which a narrow, heavily reduced $TiO_{2-x}$ sublayer acted as an oxygen-deficient, electronically conductive region and the adjacent stoichiometric $TiO_2$ layer acted as a resistive barrier. Under an applied voltage, the oxygen vacancies populating the sublayer drift along the electric field, and the boundary between the conductive and resistive regions is displaced through the film. Because the overall resistance of the stack is determined by the geometric proportion of the two regions, the device resistance is modulated continuously by the displacement of that internal boundary. In the language of eq. (1.1), the position of the boundary plays the role of the internal state $\mathbf{x}$, the drift of oxygen vacancies provides the state equation $f$, and the pinched hysteresis predicted by Chua's theory is recovered quantitatively from the resulting $i$–$v$ response.

The immediate impact of this result on the broader research community was disproportionate to the incremental physics it reported. By identifying an explicit, straightforwardly fabricated thin-film device with a pre-existing and, until then, largely formal theoretical construct, Strukov and co-workers converted the memristor from a mathematical curiosity into a concrete experimental object, and in doing so catalysed a large and still expanding body of work into resistive-switching and memristive devices across the major electronic-materials communities. The fact that the resistance of a two-terminal device could be set to a quasi-continuous range of values by applied electrical pulses, and read non-destructively at low voltage, was immediately recognised as a physical realisation of the analogue weight element required for neuromorphic hardware [39]. Within a few years, demonstrations of crossbar arrays of such devices operating as in-memory analogue matrix–vector multipliers had appeared [9, 40], and the memristor had become the template against which a wide variety of subsequent device classes — including the organic polymer-composite devices studied in this thesis — are now routinely compared. It is important, however, to register a second lesson from the Strukov paper that is often overlooked: the 2008 $TiO_2$ device is a specific instance of one specific resistive-switching mechanism, namely the drift of oxygen vacancies in a transition-metal oxide. The generality of the memristive formalism is considerably greater than that particular realisation, and a responsible survey of the field must recognise several mechanistically distinct families that operate under the same umbrella.

### 1.3.3 Classification of Memristive Mechanisms

The term *memristor*, as currently used in the device-physics literature, designates a phenomenological category rather than a microscopic mechanism. A device is called memristive if its current–voltage characteristic exhibits the pinched hysteresis loop predicted by Chua's theory and if its internal state can be set and read electrically; it is not required to rely on any specific physical process to do so. Review articles have accordingly identified several broad mechanistic families [39, 40], each of which satisfies the memristive definition through a different physical mechanism and each of which exhibits a different profile of operational strengths and weaknesses. A brief taxonomy of these families is instructive, both because it clarifies the physical diversity hidden behind the umbrella term and because it situates the ionic-polymer devices studied in this thesis within the broader experimental landscape.

The first and historically most visible family is *filamentary* resistive switching. Under an applied voltage, a narrow conductive path — a filament — is formed across an otherwise insulating thin film, and is subsequently ruptured or reformed by further electrical stimuli. In *conductive-bridge* random-access memories (CBRAM), the filament is composed of metal atoms electrochemically displaced from an active electrode (typically silver or copper) through a solid electrolyte. In *valence-change* memories, often based on binary transition-metal oxides, the filament is composed of ordered oxygen vacancies or partially reduced metal-cation states within the oxide lattice [41]. Filamentary devices typically exhibit high ON/OFF ratios, short switching times, and non-volatile retention; their principal drawbacks are the stochastic nature of filament nucleation and dissolution, which translates into substantial device-to-device and cycle-to-cycle variability [42], and the restricted number of reliably distinguishable intermediate conductance states that this stochasticity imposes. Filamentary devices are therefore well suited to binary storage and to coarse multi-level weights, but they are a less natural platform for dense analogue representations.

A second family operates through *interface* or *non-filamentary* switching, in which the active area of the device modulates its conductance uniformly rather than through a localised filament. Typical implementations modulate the height or width of a Schottky barrier at a metal–semiconductor junction through the accumulation or depletion of mobile ionic species near the interface, or, in certain mixed ionic–electronic systems, through a distributed electrochemical doping of the bulk semiconductor [12, 13]. Because the modulation is spatially distributed rather than localised, interface-type devices usually support a quasi-continuous range of intermediate conductance values and therefore finer analogue gradation than their filamentary counterparts. Their ON/OFF ratios tend to be smaller, but their variability is typically lower and their current scales predictably with the junction area, which simplifies array integration. Several of the organic electrochemical devices surveyed later in this chapter fall into this class, and so, in large part, do the ionic polymer-composite devices on which the experimental work of this thesis is based.

A third family consists of *phase-change* memories, in which the active element is a chalcogenide alloy (most commonly $Ge_2Sb_2Te_5$ or closely related compositions) whose electrical resistance depends on whether it is in a crystalline or amorphous phase. Programming is performed by Joule-heating pulses that melt and rapidly quench, or partially anneal, the material [10, 11, 43]. Phase-change memories typically achieve excellent retention, large ON/OFF ratios, and high endurance, at the cost of substantial per-write energies and a tendency towards binary or few-state behaviour unless specifically engineered to support analogue intermediate states. They have become a mature technology for storage-class-memory applications and are increasingly deployed in in-memory analogue-computing demonstrations.

A fourth family, the *spintronic* or magnetic resistive device, implements memristive behaviour through the magnetic configuration of a tunnel junction whose resistance depends on the relative orientation of two ferromagnetic layers [44]. Spintronic devices operate at very high speed and low per-write energy, and they are the basis of commercial spin-transfer-torque magnetic random-access memory. Their principal limitations for neuromorphic applications are a relatively small ON/OFF ratio (often of order two) and a limited range of stable intermediate states, although recent work has shown that their intrinsic stochastic dynamics can itself be exploited as a computational resource within appropriately designed architectures [44, 45].

A fifth family, and the one of direct relevance to this thesis, is that of *ionic* memristors, in which the internal state is the distribution of one or more mobile ionic species within a solid matrix. Ionic memristors can be further subdivided by the nature of the host matrix — inorganic oxide, chalcogenide, mixed ionic–electronic conductor, or polymer electrolyte — and by the identity of the mobile species, which range from small cations such as $Li^+$, $Na^+$, or $Ag^+$ to protons and a variety of small anions. What distinguishes ionic memristors as a class is that the internal state is a chemical configuration of mobile species, not a morphological or magnetic configuration of the host, and that the kinetics of that state are governed by ionic transport and binding within the matrix rather than by electronic processes alone. Two consequences follow that bear directly on the subsequent argument of this thesis. First, ionic systems exhibit intrinsically history-dependent dynamics, because the distribution of mobile species at any given instant reflects the cumulative effect of past electric fields on their migration. Second, and equally important, ionic systems typically exhibit spontaneous *relaxation* in the absence of a driving field, because the same intermolecular forces and entropic gradients that oppose the non-equilibrium distribution imposed under bias drive the system back towards equilibrium once the bias is removed. Volatility is therefore not a pathology of ionic memristors but an intrinsic feature of the underlying physics, and its timescales are set by the kinetics of ionic transport, which are in turn chemically tunable through the composition of the host matrix. This fifth family, and in particular its organic polymer-electrolyte subfamily, is the platform on which the experimental work of this thesis is built.

### 1.3.4 Key Memristive Parameters

The mechanistic heterogeneity surveyed in section 1.3.3 is reflected in the specialised vocabulary that has grown up around the evaluation of memristive devices. A typical specification report will quote several parameters. The *high-resistance state* (HRS) and *low-resistance state* (LRS) define the extremes of the accessible conductance range; their ratio is the *ON/OFF ratio*, a figure of merit for the dynamic range of the device. The *set* and *reset voltages* specify the magnitude of electrical stimuli required to program the device between its extreme states, and the associated *write energy* sets the per-operation energetic cost. The *switching speed* describes how rapidly the device responds to a programming pulse. *Retention* quantifies the time for which a given state is preserved without re-programming, and *endurance* the number of programming cycles the device can support before its behaviour degrades. Parameters relevant specifically to analogue operation include the *linearity* and *symmetry* of the conductance update under identical positive and negative pulses, and the *device-to-device* and *cycle-to-cycle variability* of all of the above.

These parameters have their natural setting in the evaluation of devices intended for *trained-weight* storage in analogue in-memory-computing arrays [9, 40]. In that setting, the target application is a fixed conductance matrix implementing a trained neural network; retention must be long enough to preserve trained weights between re-programming events, typically of order seconds to years depending on the deployment scenario; endurance must support occasional re-training and periodic write–verify cycles; linearity and symmetry are central to the numerical accuracy of gradient-based training performed in place on the array; and variability translates directly into a loss of inference accuracy at deployment. A large fraction of the memristor-device literature is written from this standpoint, and it is one of the principal contexts in which organic and polymer-composite devices have been judged — frequently unfavourably, because their retention times are short, their endurance is limited, and their device-to-device variability is high.

A more careful reading of the same parameter list, however, reveals that not all of its entries are unambiguously desirable when their values are maximised. Retention is the most immediate example. For a trained-weight device, a short retention time is strictly a defect, because it causes the encoded weight to decay between inference operations. For a fading-memory device embedded in a physical-reservoir or event-driven architecture, by contrast, the retention time *is* the computational parameter that sets the effective temporal window over which the device integrates past inputs. A retention that is too long renders the device insensitive to temporal structure on the timescale of the incoming events; a retention that is too short fails to resolve correlations between events that are not strictly simultaneous. There is therefore a task-dependent optimum for the retention, and the engineering desideratum is not "as long as possible" but "matched to the signal timescale". An analogous shift of interpretation applies to switching speed, write energy, linearity, and symmetry: the values that are optimal for a trained-weight element differ, sometimes substantially, from those that are optimal for a fading-memory element, and a single device cannot, in general, simultaneously optimise both sets.

*Variability* occupies a more delicate position in this reframing, and its treatment requires care. Device-to-device and cycle-to-cycle variability is, in the trained-weight regime, a defect: it contributes to the loss of inference accuracy whenever the actual conductance differs from the trained target, and its mitigation is a central concern in the design of in-memory-computing hardware. In the reservoir-computing regime, by contrast, the relevant quantity is not the fidelity with which an individual device reproduces a target conductance but the *diversity* of the collection of devices taken as a whole. A reservoir constructed from elements with a spread of intrinsic time constants and nonlinear transfer characteristics is, all else being equal, a richer computational substrate than a reservoir constructed from a hypothetical set of identical elements, because the spread provides independent nonlinear projections of the input onto different axes of the readout space [22]. This observation must, however, be stated carefully. It is not a claim that any amount of variability is beneficial: uncontrolled variability within a single device, without a stable underlying dynamical structure, remains a defect regardless of architecture. It is rather a claim that *inter-device heterogeneity*, provided that each individual device is reproducible within reasonable bounds and that its dynamical response is well characterised, should be regarded as a resource rather than a failure mode. The distinction between intra-device instability, which is a defect, and inter-device heterogeneity, which is a resource, is subtle but important, and it will recur throughout the evaluation of the polymer-composite devices reported in the later chapters of this thesis.

### 1.3.5 Dynamic Computing Paradigms for Volatile Memristors

The reframing of memristive parameters sketched in section 1.3.4 is not a cosmetic relabelling: it reflects the emergence, over the past decade, of a set of computational paradigms specifically adapted to volatile, heterogeneous, history-dependent devices. Within these paradigms, a memristor whose state decays on a well-defined timescale is not a poor substitute for a non-volatile memory but the natural physical instantiation of a *leaky integrator*, the elementary dynamical building block of temporal computation.

The computational meaning of such a device is fixed not by how long it can hold a given value but by what it does to an incoming temporal signal: how it smooths it, how it emphasises certain frequencies or inter-event intervals, how it transforms it into a representation on which a downstream linear readout can then operate. The three paradigms surveyed below will be revisited in greater technical detail in a later chapter of this thesis, where the polymer-electrolyte devices under study will be evaluated against exactly the metrics that these paradigms imply, rather than against the trained-weight specifications introduced in sections 1.2.3 and 1.3.4.

The most prominent of the three paradigms is *physical reservoir computing* [22], introduced at the conceptual level in section 1.1.2. In a volatile-memristor reservoir, the input signal is encoded as a stream of voltage pulses presented to an array of memristive devices whose internal states evolve under the applied stimuli and relax between them. At any instant, the collective state of the array provides a nonlinear projection of the recent history of the input onto a high-dimensional feature space, and a linear readout trained by ordinary least squares on this space is sufficient to solve a broad class of time-varying classification, regression, and prediction tasks. Experimental demonstrations of this principle in inorganic volatile memristors have been reported for diffusive metal-filament devices [23, 24] and dynamic oxide memristors [25], and demonstrations in organic electrochemical systems have begun to appear [26]. In all of these cases, the short retention of the individual device is not a limitation of the method but its enabling physical property: it is the retention that supplies each device with a well-defined fading-memory time constant.

A second paradigm exploits volatile memristors as *coincidence detectors* and *temporal-correlation extractors*. Because the internal state of such a device relaxes in the absence of a driving pulse, a pulse train whose inter-pulse intervals are short compared with the relaxation timescale produces a progressive accumulation of state, while a train whose intervals are long produces essentially no accumulation. The integrated response of the device therefore carries direct information about the temporal proximity of successive events, and can be used to discriminate coincident or near-coincident inputs from temporally dispersed ones. Architectures based on this principle have been proposed and partially demonstrated in combination with STDP-like plasticity rules, in which the sign and magnitude of the long-term response depend on the relative timing of pre- and post-synaptic events [46]. The approach constitutes, in effect, a direct hardware transcription of the spike-timing-sensitive plasticity introduced in section 1.2.2, and it relies essentially on the relaxation dynamics provided by the volatile memristor.

A third paradigm employs banks of volatile memristors, engineered to exhibit distinct relaxation time constants, as *multi-timescale transient filter banks* for streaming signals. In such an architecture, a single input is processed in parallel by devices of different intrinsic timescales: elements with short time constants respond sharply to fast transients and return quickly to baseline, while elements with long time constants respond more gradually and integrate over longer windows. The ensemble of outputs is then read by a downstream classifier to detect onsets, bursts, rhythmic patterns, or slow envelopes that could not be extracted from any single time constant acting in isolation. The chemical tunability of ionic memristors, and in particular their capacity to span several orders of magnitude in relaxation timescale through modest changes in host composition or mobile-species identity, makes them especially well matched to this architecture.

The performance metrics relevant to these three paradigms differ systematically from the trained-weight metrics catalogued in section 1.3.4. For reservoir computing, the principal figure of merit is the *memory capacity* of the device ensemble, a quantitative measure of how faithfully the reservoir state reconstructs past inputs at various lag times, together with the *class separability* it provides on a representative benchmark task and its *robustness* to the intrinsic variability of the constituent devices [47]. For coincidence detection and temporal-correlation extraction, the relevant metrics include the width and symmetry of the effective timing window, the discrimination margin between coincident and dispersed inputs, and the energy cost per detected event. For multi-timescale filtering, the central quantities are the span of accessible time constants across the ensemble, the linearity and isolation of each channel, and the system-level power consumption under realistic input statistics. None of these metrics is meaningfully captured by HRS, LRS, ON/OFF ratio, or retention in isolation. They share, instead, a common conceptual feature: they are *task-matched*, and they reward a device whose physical timescales are aligned with the temporal structure of the signals that the device is required to process.

The argument of this section can therefore be summarised as follows. The memristor is simultaneously a precise concept in circuit theory, a heterogeneous family of physical devices, and a conceptual bridge between the synaptic template developed in section 1.2 and a set of emerging computational paradigms that exploit history-dependent device dynamics. Within the memristive-systems framework formalised by Chua and Kang, both persistent-state and relaxing-state devices are first-class objects, and each class has its own natural computational role; the choice between the two is not a hierarchy in which one is superior and the other a pale imitation, but a match between device physics and computational target. Among the five mechanistic families surveyed above, ionic memristors stand out for the present work because their internal state is, by construction, a chemical configuration of mobile species, and because their relaxation dynamics and history dependence are intrinsic rather than engineered around. The question that the next section will address is the natural one: among the many classes of ionic memristors described in the literature, which material platform offers the combination of chemical tunability, processing versatility, and operational robustness required for the kind of analogue, multi-timescale dynamics that this thesis pursues? The answer, as will be argued in detail in the sections that follow, points away from inorganic oxides and towards organic and polymeric hosts, in which the relevant transport and binding chemistries are directly accessible to molecular design and the device physics can be tuned systematically through composition.

## 1.4 Inorganic Memristors for Neuromorphic Computing

The classification exercise of section 1.3.3 catalogued the principal mechanistic families through which memristive behaviour has been implemented at the device level. The survey presented in that subsection was, however, deliberately abstract: its purpose was to establish the physical diversity of the memristor concept rather than to report on the technological state of any particular family. The present section narrows the discussion to the three inorganic families that, taken together, account for the bulk of present-day neuromorphic-hardware activity — metal-oxide resistive memories, phase-change memories, and spin-transfer-torque magnetic tunnel junctions — and examines what each family offers to a neuromorphic designer, what each family costs, and where each family fits within the design-space map drawn in sections 1.3.4 and 1.3.5. None of these families, it will be argued, has been shown to fail in any global sense; each of them, on the contrary, occupies a distinct and technologically powerful region of the design space. The conclusion of the section is correspondingly measured: inorganic memristors have firmly established the credibility of memristive hardware as a post-von-Neumann substrate, but the specific combination of properties that this thesis pursues — analogue, ion-mediated, chemically tunable, timescale-rich behaviour on solution-processable soft substrates — lies outside the regions in which these inorganic families are naturally strong, and motivates a complementary investigation in organic and polymeric materials. The present section makes this case explicitly, and in a form that avoids the common rhetorical trap of contrasting organic devices against an inorganic strawman.

### 1.4.1 Metal-Oxide Memristors

The single broadest and most technologically developed inorganic memristor family operates through resistive switching in transition-metal-oxide thin films. Following the identification of the $TiO_2$ device of Strukov and co-workers [38] with Chua's memristor concept, research activity shifted rapidly to a wider palette of oxide systems, of which the best-studied are hafnium dioxide ($HfO_2$), tantalum oxide ($TaO_x$), nickel oxide (NiO), tungsten oxide ($WO_x$), and titanium dioxide itself. Many of these materials had been used for years as gate-dielectric or interconnect-related layers in conventional silicon technology, and their integration into back-end-of-line processes was therefore immediately feasible. This compatibility with standard CMOS flows is the most important practical reason for the prominence of metal-oxide memristors in neuromorphic-hardware demonstrations: arrays of thousands to millions of such devices can be fabricated on top of existing addressing electronics without disruption of the underlying silicon, and large-scale integrated demonstrations — including the metal-oxide crossbar of Prezioso et al. [9] — have followed accordingly.

At the device level, metal-oxide memristors are usually classified mechanistically as either *valence-change memories* (VCM) or *electrochemical metallisation / conductive-bridge* memories (CBRAM), according to which mobile species carries the switching behaviour. In VCM devices, the mobile species is an oxygen vacancy, whose field-driven migration produces local reduction and oxidation of the host cation and a corresponding change in the electronic conductance of a confined filamentary region [41]. In CBRAM devices, the mobile species is a metal cation originating from an electrochemically active electrode (typically $Ag^+$ or $Cu^+$), which migrates across the oxide layer under the applied field and plates out into a metallic filament that shorts the stack. Both mechanisms commonly require an initial *forming* step, in which a localised dielectric soft breakdown establishes the filament geometry that subsequent set and reset pulses can then modulate. Once formed, the device exhibits a bistable or, depending on the compliance current and pulse protocol, a graded multi-state resistance. A third, intermediate class of metal-oxide devices operates through non-filamentary *interface-type* modulation, in which the conductance of the entire device area is adjusted through distributed oxygen-vacancy redistribution at a metal–oxide Schottky interface, and which tends to support finer analogue gradation at the cost of lower ON/OFF ratios.

The strengths of metal-oxide memristors are, in outline, those of a mature binary- or few-level non-volatile memory. Retention times of years at room temperature are routinely reported; switching times can reach the nanosecond range; endurance can exceed $10^9$ cycles in the best-engineered stacks; and densities compatible with sub-10 nm crossbars are in principle achievable [39, 40]. These properties have made metal-oxide resistive memory the canonical platform for commercial and pre-commercial in-memory computing demonstrations, and the one against which most other families are implicitly benchmarked. Volatile subspecies of the same family are known as well: diffusive metal-filament devices, in which the filament spontaneously disrupts by Ag or Cu nanocluster diffusion after the programming bias is removed, have been used explicitly as fading-memory reservoir elements [23–25]. Metal-oxide memristors are therefore not a uniformly non-volatile family, and the framing used in this thesis does not treat them as such; the literature instead supports a designer's choice of persistent or relaxing behaviour through controlled choice of materials, stack geometry, and programming protocol.

The weaknesses of the family for high-precision analogue synaptic operation, however, are equally well documented. The conductance modulation in filamentary devices is intrinsically stochastic: the nucleation of the filament, its lateral position within the active area, and the exact cross-section through which it conducts are determined by thermally and electrically driven fluctuations at the nanoscale, and even in carefully engineered stacks a substantial device-to-device and cycle-to-cycle spread in set voltage, reset voltage, and achieved conductance remains [42]. The associated non-ideal linearity and asymmetry of the conductance update under identical positive and negative pulses have been repeatedly identified as limiting factors in the numerical accuracy of in-place gradient-based training on metal-oxide crossbars [40]. Multi-level operation is possible but is usually restricted, in practice, to a small number of reliably distinguishable states; obtaining a genuinely high-resolution analogue weight from a single filamentary cell remains difficult, and common mitigation strategies rely on multi-cell unit codes or on hybrid digital–analogue representations. None of these observations denies the engineering maturity of the metal-oxide platform. They do however clarify that the platform is optimised for a region of the design space in which retention, endurance, and integration density are maximised, and in which analogue linearity, symmetric bidirectional update, and fine-grained timescale engineering are secondary concerns.

### 1.4.2 Phase-Change Memory

A conceptually distinct inorganic family operates through reversible switching between the amorphous and crystalline phases of a chalcogenide alloy. The canonical alloy is $Ge_2Sb_2Te_5$, often abbreviated GST, and a wide range of closely related compositions (doped variants of GST, Ag−In−Sb−Te, $Sb_2Te$, and so on) exhibit the same essential switching physics. In the crystalline phase, the alloy is an electronically conductive narrow-gap semiconductor; in the amorphous phase it is effectively insulating, with resistance several orders of magnitude higher than the crystalline value. The transition between the two phases is driven thermally: a short, intense *reset* pulse melts a small volume of the alloy and is terminated rapidly enough for the melt to vitrify into the amorphous state, while a longer, lower-amplitude *set* pulse holds the material at a temperature above the crystallisation threshold but below the melting point, allowing the disordered region to re-crystallise [10, 11]. The resulting resistance is not, as in filamentary metal-oxide memristors, the property of a localised channel but of the geometric proportion of crystalline and amorphous phases within the programmed volume.

The technological maturity of phase-change memory is unusually high among memristive device families. It has been under sustained development as a storage-class memory since the 1970s, has been the subject of multiple commercial product generations, and has, over the past decade, become the most credible inorganic platform for analogue in-memory computing. Le Gallo and co-workers demonstrated mixed-precision in-memory arithmetic on PCM arrays with sufficient accuracy to support, jointly with a digital co-processor, the solution of linear systems of practical scale [43]. Ambrogio et al. [48] subsequently demonstrated equivalent-accuracy training of modest-scale neural networks using PCM elements as the long-term weight store, paired with a volatile capacitor for fine-grained weight updates. These demonstrations, taken together, establish PCM as the most seriously deployed analogue-memristive substrate for conventional deep-learning workloads, and the benchmark against which any alternative analogue weight element must be evaluated for trained-weight applications.

The constraints of phase-change memory, however, are significant when the device is viewed as a candidate synapse rather than as a weight cell. The reset operation requires local melting of the active volume, and the associated current densities and thermal loads impose both a high per-write energy and a stringent requirement on thermal isolation of neighbouring cells. The amorphous phase produced by quenching is, moreover, not perfectly stable: its resistance drifts over time according to an approximately power-law relaxation of the amorphous structure, typically of order a few per cent of the baseline per decade in time, which must be compensated by periodic re-reading or by read-time calibration. Precise analogue programming of intermediate resistance states relies on carefully shaped pulse protocols and on tight thermal control, and is therefore more sensitive to process and operating-condition variation than the corresponding bistable operation. The underlying physics is also, by construction, thermal and structural, which leaves only limited room for chemical tuning of the key timescales: the crystallisation and amorphisation kinetics are set by the alloy composition and the device geometry, both of which can be adjusted within modest bounds but none of which provides the kind of molecular-level handle on ion transport that the soft-matter systems discussed in the next section offer. PCM is, in summary, the most compelling analogue in-memory-computing platform currently available, but its region of strength is the high-retention, high-endurance, trained-weight corner of the design space, not the fast, soft, chemically tunable, fading-memory corner that this thesis pursues.

### 1.4.3 Spin-Transfer-Torque Magnetic Tunnel Junctions

The third inorganic memristor family departs more sharply from the ion- or atom-rearrangement mechanisms discussed so far. A magnetic tunnel junction (MTJ) consists of two ferromagnetic layers separated by a thin insulating tunnel barrier (typically MgO). The resistance of the junction depends, through the tunnelling magnetoresistance (TMR) effect, on the relative orientation of the magnetisations of the two layers: it is low when the magnetisations are parallel and high when they are antiparallel. In a spin-transfer-torque (STT) device, the free layer can be switched between the parallel and antiparallel configurations by a spin-polarised tunnel current: conduction electrons traversing the barrier from the fixed reference layer acquire a net spin polarisation, and their subsequent interaction with the free-layer moment exerts a torque sufficient, above a threshold current density, to reverse the free layer. Read and write operations therefore share the same electrical port, and the internal state variable is the magnetic configuration of the free layer rather than a chemical or morphological property of the host material.

The STT–MTJ family is of considerable interest for neuromorphic and unconventional computing for several reasons. Switching times in the nanosecond range, endurance exceeding $10^{12}$ cycles, write energies per bit in the femtojoule range, and compatibility with back-end-of-line CMOS integration combine to make the device attractive as a low-energy non-volatile element. The magnetic origin of the switching threshold, combined with thermal fluctuations of the free layer in devices close to the superparamagnetic limit, also offers a natural and well-controlled source of stochasticity, which has been exploited in probabilistic-computing and neuromorphic demonstrations: Torrejon et al. [44] used spin-torque nano-oscillators as dynamical nodes for physical reservoir computing and achieved competitive accuracy on a spoken-digit classification benchmark, and Mizrahi et al. [45] used populations of superparamagnetic MTJs as noisy basis functions for neural-style computation. These demonstrations show that the MTJ family is not restricted to digital memory operation and that its dynamical behaviour can itself be a computational resource.

The limitations of the platform, nonetheless, are real in the context of fine-grained analogue synaptic operation. A simple STT–MTJ is essentially a two-state device, with the ratio of antiparallel to parallel resistance typically in the range of two to a few; extending this ratio and introducing a larger number of reliably programmable intermediate states requires more elaborate stack engineering (synthetic antiferromagnetic references, perpendicular-anisotropy stacks, or multi-layer configurations) and tighter process control, with corresponding penalties in yield and integration complexity. The thermal stability of the free layer, which sets the retention, is in direct tension with the switching current, which sets the write energy, and trading one against the other imposes a tight envelope within which the device must operate. Finally, the materials involved — carefully engineered multilayers of ferromagnetic metals, synthetic antiferromagnets, and crystalline MgO tunnel barriers — are demanding in their growth requirements and do not lend themselves to the kind of solution-processable, substrate-agnostic fabrication that is straightforward for soft-matter devices. STT–MTJs are therefore best thought of as fast, non-volatile, few-state, stochastic elements whose strengths lie in probabilistic computing and in the replacement of small embedded non-volatile memories, rather than as a natural substrate for analogue, multi-timescale synaptic emulation.

### 1.4.4 General Limitations of Inorganic Memristors

The three families surveyed above differ substantially in their mechanisms, their strengths, and the regions of the design space in which they are most naturally deployed. They nevertheless share a set of features that, taken together, constrain their collective coverage of the neuromorphic design space and that motivate the complementary exploration pursued in the remainder of this thesis. These shared features are emphasised here not to argue that inorganic memristors have failed — the preceding subsections make clear that they have not — but to clarify where the inorganic families are strong, where they are less natural, and why a materials-space investigation beyond them is scientifically warranted.

The first shared constraint is a limited range of *chemical tunability*. Each of the three families is anchored in a small number of canonical host materials — the transition-metal oxides, the tellurium-based chalcogenide alloys, and a restricted set of engineered ferromagnetic stacks — whose switching behaviour is determined by the intrinsic physics of these hosts. The available design knobs within each family are primarily geometric (layer thicknesses, active area, electrode contacts), stoichiometric (within narrow compositional windows), or process-related (annealing conditions, forming protocols). Access to the switching physics itself is therefore mediated, in practice, through a handful of well-characterised stacks; it is rarely possible to modify the key energetic or kinetic parameters of the device by, for example, substituting one mobile species for another with different binding chemistry, or by exchanging the host polymer for a chemically distinct analogue with a different glass-transition temperature. This is a statement about the structure of the design space rather than a criticism of any individual demonstration; molecular-level tunability of the binding and transport chemistry simply is not among the handles that the inorganic platforms expose.

A second and closely related constraint concerns *dynamical timescale engineering*. Each of the three inorganic families admits both non-volatile and, in specific implementations, volatile behaviour: diffusive metal-filament devices in the metal-oxide family supply fading-memory dynamics [23–25, 49], resistance drift in the amorphous phase of PCM provides a (less well-controlled) relaxation [11], and superparamagnetic MTJs introduce stochastic fluctuations on engineered timescales [45]. What is more difficult, in each case, is to *systematically* sweep a relaxation time constant over several orders of magnitude within a single materials platform by way of a single chemical variable. The underlying kinetic processes — vacancy drift in a crystalline oxide, thermally activated crystallisation in a chalcogenide melt, or magnetic relaxation of a ferromagnetic free layer — are determined by the intrinsic energy landscape of the host, and the span of accessible timescales within each family is correspondingly constrained. For applications such as the heterogeneous reservoir or the multi-timescale transient filter bank described in section 1.3.5, in which parallel devices with deliberately distinct time constants are required, this constraint is not peripheral but central.

A third constraint concerns *fabrication and integration*. The inorganic families above rely, with few exceptions, on vacuum-based thin-film deposition, on high-temperature anneals, and on top-down lithographic patterning. These workflows are fully compatible with conventional silicon technology, which is an unambiguous strength in the context of dense crossbar arrays integrated on CMOS, but they constrain the substrates on which the devices can be fabricated and the operating temperatures at which they can be processed. Large-area, flexible, biocompatible, or printable platforms — relevant for soft-electronic, implantable, or distributed-sensor neuromorphic systems — are not the natural home of these workflows, and adapting them to such contexts requires substantial process engineering. Solution-processable active layers, by contrast, can in principle be deposited on arbitrary substrates at mild temperatures, and their active chemistry can be tuned by varying the composition of the precursor solution.

This is a feature of the fabrication physics, not a deficiency of any specific inorganic demonstration, but it remains a feature of first-order practical importance for the class of applications considered in the later chapters of this thesis.

A fourth constraint concerns the mechanistic *stochasticity* that is inherent to filamentary and phase-change operation. In the context of trained-weight storage, stochasticity is a defect whose mitigation dominates a large fraction of the device-engineering effort. In the reservoir-computing regime, by contrast, stochasticity can contribute usefully to the nonlinear diversity of the reservoir, provided that it is controlled at the level of *inter-device heterogeneity* and not of *intra-device instability*, in the sense introduced in section 1.3.4. The inorganic families above tend to exhibit both; disentangling the two is possible but requires considerable process control. A device class in which inter-device heterogeneity can be introduced deliberately through composition, while intra-device stability is preserved by the collective character of the conduction mechanism, would therefore offer a cleaner route to the controlled-variability operating regime that the fading-memory paradigms benefit from.

Taken together, these four constraints converge on a single observation. The inorganic memristor families convincingly validate the broader memristive-neuromorphic route: they have demonstrated, at the level of operating hardware, that analogue, multi-state, history-dependent electronic devices can be built and integrated at scale, and that they can support competitive in-memory arithmetic and reservoir-style temporal computation. What they leave unresolved is a materials-space question. The combination of analogue, ion-mediated, chemically tunable, and timescale-rich behaviour on soft, solution-processable platforms is not a region in which the inorganic families are most naturally strong; it is a region in which they are in principle accessible but practically constrained by their intrinsic physics and by their fabrication workflows. The question that the next section will pose is therefore the natural complement of the one that opened this one: is there a class of materials in which the active ionic species, their binding environment, and their kinetics are directly accessible to molecular design, and in which the key memristive timescales can be engineered through straightforward compositional variation rather than through repeated redesign of rigid thin-film stacks? The answer, as will be argued in detail in the sections that follow, is that organic and polymeric ion-mediated materials offer precisely this combination of properties, and that they do so in a way that is complementary to, rather than competitive with, the established strengths of the inorganic families.

## 1.5 Organic Materials for Memristive Applications

The preceding section closed on a materials-space question. The inorganic memristor families validate the broader memristive and neuromorphic programme, and each of them occupies a region of the design space in which it is genuinely strong, but they share a restricted palette of chemical handles and a workflow that is most natural on rigid, vacuum-processed, high-temperature-compatible substrates. The combination of analogue multi-state operation, ion-mediated internal dynamics, controllable timescales across several orders of magnitude, and compatibility with soft, conformable, or biointerfaced form factors is not a region in which they are most naturally located. Organic materials are attractive precisely where that combination is required. The attractiveness is not driven by a general-purpose claim that organic devices are cheaper, cleaner, or more reproducible than their inorganic counterparts — claims of that kind are often either unsupported or highly context-dependent — but by a specific structural fact: in organic and polymeric systems, the chemistry that sets the electronic transport, the ionic transport, the redox activity, and the relaxation kinetics is directly accessible to molecular design. Tuning any of these properties is a matter of synthesis and formulation rather than of repeated redesign of a rigid thin-film stack, and the resulting changes in device dynamics can be mapped, in principle, onto the underlying chemistry in a systematic way.

This section develops that argument at field level. section 1.5.1 identifies the features that genuinely distinguish the organic materials class from inorganic platforms for memristive and neuromorphic applications, and states carefully what those features do and do not guarantee. section 1.5.2 surveys the main physical mechanisms that have been invoked to explain organic memristive behaviour, clarifies that they do not form a single mechanism class, and gives particular weight to ion-mediated and mixed ionic–electronic operation, which is the route most directly relevant to the remainder of this thesis. section 1.5.3 provides a concise historical survey of the field from the early electrochemical and polymer-film demonstrations through the three-terminal electrochemical-transistor synapses, the two-terminal redox-based biomimetic devices, and the stretchable and molecular implementations of the last decade, with the aim of identifying what the field has progressively solved and which combinations of features were still unresolved at the point where the present thesis was initiated. The section closes on the specific unresolved question that motivates the next one: within the organic memristor landscape, which architecture is most suitable for controllable, reversible, ion-mediated neuromorphic dynamics?

### 1.5.1 Advantages of Organic Memristors

The distinctive feature of organic materials, in the context of memristive and neuromorphic device engineering, is the breadth of chemical freedom available at the level of the active layer. A semiconducting polymer can be designed by choice of backbone, side-chain functionalisation, and molecular weight distribution; an ion-transport host can be tuned by choice of chelating motif, polymer topology, and glass-transition temperature; a redox-active moiety can be selected from an enormous library of molecular and polymeric species with characterised oxidation potentials and known redox reversibilities; a mobile ionic species can be chosen independently of the host on the basis of charge, radius, and Lewis acidity. The relevant point is not that any individual organic device is automatically superior to an inorganic equivalent, but that the number of orthogonal chemical handles available for engineering its internal-state physics is considerably larger. In particular, the ionic component of the active layer — the identity of the mobile cation and anion, the nature of the solvating environment, and the segmental dynamics of the host polymer — is directly under synthetic and formulation control in a way that is not available, at the equivalent level of granularity, in the inorganic platforms discussed in section 1.4.

Solution processability is the second feature that genuinely distinguishes the organic class. A large fraction of organic active layers relevant for memristive operation — conjugated polymers, hyperbranched polyethers, polyelectrolytes, redox-active polymers, and their blends — can be deposited from solution by techniques ranging from spin coating and blade coating in the laboratory to inkjet, spray, and roll-to-roll processes on larger areas. This is a feature of the fabrication physics, not a promise of low cost or of industrial scalability per se, and its practical benefits depend on the specific material and solvent system used. What it does guarantee is that the active layer composition can be varied continuously, by mixing precursor solutions in a single step, which is the natural mode of fabrication for composite devices in which an electronic component and an ionic component coexist. The ability to adjust the polymer-to-semiconductor and salt-to-semiconductor ratios simply by changing the solution recipe, rather than by redesigning a multilayer stack, is the fabrication-level counterpart of the chemical freedom described above.

A closely related feature is the mild thermal budget required for processing. Most organic memristive layers are fabricated at temperatures below one hundred and fifty degrees Celsius, frequently below one hundred, and the top contacts can be deposited through shadow masks by thermal evaporation without high-temperature post-anneals. This is compatible with flexible plastic substrates, paper, textiles, elastomers, and biological tissues, and it places no strict lower bound on the substrate thickness beyond the mechanical requirements of handling. The corresponding properties of the finished device — mechanical flexibility under bending, conformability over curved surfaces, and in some cases stretchability — follow naturally from this processing regime rather than from any intrinsic mechanical property of a single active molecule. Demonstrations of flexible and stretchable organic synaptic devices [50, 51] illustrate that mechanical compliance is accessible at the device level with standard organic-electronic fabrication, which is relevant for wearable, implantable, or conformal neuromorphic systems of the kind envisaged under the *nervetronic* [51] heading.

These three features — chemical freedom, solution processability, and mild thermal budget — together describe a design space that is qualitatively different from the inorganic one. Nothing in this description guarantees that an arbitrary organic memristor will be cheap, green, reproducible, biocompatible, or long-lived. Each of those properties depends on choices about specific materials, solvents, contact metals, encapsulation strategies, and operating conditions, and blanket claims to that effect are often not supported in the primary literature. What can be claimed, and what is enough for the argument pursued here, is that the organic class exposes handles on the internal-state physics of the device that the inorganic class does not, and that those handles operate through chemistry rather than through the re-engineering of a rigid thin-film stack.

Two consequences of this design space are of particular importance for the thesis-wide argument developed in sections 1.3 and 1.4. The first is that richly tunable analogue responses are readily achievable in organic systems, because the microscopic mechanisms underlying the switching behaviour — charge trapping on molecular sites, redox activity of a macromolecular host, displacement of mobile ions in a polymer matrix — are typically distributed over a large number of sites and therefore admit many intermediate states rather than a sharp binary transition. The second is that the relaxation timescales of the device can be engineered by chemistry. Ion transport in a polymer electrolyte depends on the binding strength between the mobile ion and the polymer, on the segmental mobility of the host, and on the concentration of mobile species; each of these parameters is a design variable in an organic composite. Device timescales ranging from milliseconds to many seconds, matched to the timescales of biological short-term plasticity and of slow temporal signals, are therefore accessible within a single materials family by straightforward compositional changes. This is the specific feature that makes organic ion-mediated systems attractive for the volatile, fading-memory paradigms introduced in section 1.3.5, in which controllable relaxation rather than maximal retention is the relevant figure of merit.

It is important, in stating these advantages, not to collapse them into a pre-existing narrative in which all desirable synaptic properties are assumed to be achievable simultaneously. The organic materials class offers a large design space, but each concrete demonstration occupies a specific point in that space and exhibits specific strengths and specific compromises. Reversibility, endurance, state count, analogue control, and reproducibility can in principle all be addressed through chemistry, but not all of them have been addressed in any single system, and the historical development of the field (section 1.5.3) shows that the combination of these properties in a two-terminal organic device with interpretable ionic physics remained non-trivial well into the last decade. The argument of this chapter is not that organic memristors have solved the problems left open by inorganic ones, but that the organic design space contains regions — and specifically the region of ion-mediated polymer composites — in which these problems can be addressed through chemically transparent handles that are not available on the inorganic side.

### 1.5.2 Mechanisms in Organic Memristors

A recurring source of confusion in the organic memristor literature is the tacit assumption that *organic memristor* designates a single mechanism class, analogous to the way *phase-change memory* or *valence-change memory* designates a well-defined mechanism within the inorganic landscape. It does not. Memristive behaviour in organic devices has been attributed, credibly, to several distinct physical processes, which differ in the internal state variable that is actually being modulated, in the intrinsic reversibility of that modulation, in the typical timescales involved, and in the degree of analogue control and reproducibility that can be achieved. A productive survey of the field therefore proceeds by identifying the main mechanism families, stating what each one offers and what each one costs, and locating the ion-mediated family within this broader landscape. That is the programme of this subsection.

The first mechanism family is charge trapping and detrapping in an insulating or weakly conducting matrix. In this family, the active layer contains a dispersed population of deep or shallow traps — associated with molecular defect sites, with embedded nanoparticles, or with the interfaces between incompatible polymer phases — and the device state is set by the occupation of those traps under an applied bias. The resistance change is produced by the modulation of the electronic transport through the matrix as the local space-charge environment is altered by the trapped carriers. This family is attractive because trapping is a generic process that is readily observed in disordered organic semiconductors, and because switching voltages can be kept modest, but it typically offers limited analogue control: the state space is often effectively binary, the dependence of conductance on trap occupation is non-monotonic in some devices, and the microscopic identity of the traps is frequently hard to pin down, which constrains the interpretability of the mechanism. Retention is determined by the depth of the traps relative to the thermal energy, and is difficult to engineer precisely from composition alone.

The second family is redox switching, in which the active layer or a component of it contains a species whose oxidation state is changed reversibly under bias. The redox-active component can be a discrete molecule embedded in a polymer host, a redox-active polymer used as the active layer itself, or a redox-active moiety covalently bound to a conjugated backbone. The device state is set by the distribution of oxidised and reduced species, which modifies the local electronic structure and therefore the conductance. Redox switching offers a clear mechanistic picture — the internal state variable is the oxidation state of a well-defined chemical species, the electrochemical potentials that drive the transitions are experimentally accessible, and in favourable cases the redox process is genuinely reversible over many cycles. Several important organic memristors operate on this principle, including the family of redox-active conjugated polymer systems [52–54] and early electrochemical demonstrations based on coordination complexes [55]. The costs, in the organic context, are twofold. First, not every redox process is cleanly reversible: side reactions, dimerisation, and slow degradation of the active species limit the endurance of many systems. Second, purely redox-based devices tend to operate as bistable or few-state systems unless the redox response is distributed over an ensemble of sites; when distributed, the analogue response is recovered, but interpretability begins to shade into the statistics of the ensemble rather than into the chemistry of an individual switch.

The third family is conformational or structural switching, in which the active layer contains a molecular species whose geometry, isomerisation state, or aggregation state can be modulated by the applied field, by local heating, or by coupled electrochemical processes. Classical examples include photochromic or electrochromic switches that couple to the electrical transport, and specifically designed bistable molecular architectures [53, 54] in which the conductance of the molecule depends on its conformation. These systems offer very high levels of molecular design control and in favourable cases very high precision in the switching behaviour, since the state is set by a chemically well-defined geometric rearrangement. The costs are that the relevant molecular synthesis is frequently demanding, that the switching dynamics are tied to the kinetics of a specific molecular rearrangement which may or may not match the timescales of interest, and that scaling the behaviour from single-molecule to device-level averages introduces its own set of issues.

The fourth family, and the one most directly relevant to the thesis, is ionic redistribution in mixed ionic–electronic systems. Here the active layer contains both an electronic conductor — typically a conjugated polymer — and a population of mobile ions carried by a suitable ionic host. Under applied bias, the mobile ions migrate and accumulate near the electrode interfaces or within specific regions of the active layer, where they modify the local electronic structure of the semiconductor by a combination of interfacial doping, electrostatic band bending, and electrochemical reduction or oxidation of the polymer. The resulting conductance change is continuous, multi-state, and tied directly to the amount and spatial distribution of displaced ionic charge. The microscopic picture is closely related to the operation of light-emitting electrochemical cells [56–62], in which ionic redistribution under a driving bias creates an in-situ doped junction, and to the operation of organic electrochemical transistors (OECTs), in which a gate-driven ion exchange between an electrolyte and a conducting polymer modulates the channel current. What distinguishes the memristive mode of this family from its LEC and

OECT counterparts is that, in a memristive two-terminal device, the driving bias itself plays the role of the gate, and the residual state of the ionic distribution after the bias is removed carries the memory.

This fourth family deserves particular conceptual weight. The state variable is physically transparent: it is the position of mobile ions in a polymer host. The dynamics of the state variable are governed by ionic transport kinetics, which depend on the binding environment provided by the polymer and on the chemistry of the ion itself. The relaxation of the state, when the bias is removed, is a genuine fading-memory process with a characteristic time constant that can be estimated from the physical chemistry of the composite, and that can be engineered by changing any of the compositional variables. In contrast to the charge-trapping and redox families, where the relaxation is controlled by trap depths or redox overpotentials that are often hard to tune systematically, the relaxation of an ion-mediated device is controlled by variables — cation identity, host polymer, salt concentration, film morphology — that are directly accessible to the materials chemist. Ionic and mixed ionic–electronic mechanisms therefore naturally connect the device state to transport kinetics, relaxation, and memory timescale, in exactly the sense developed in sections 1.3.4 and 1.3.5. The costs of operating in this regime are primarily that the ionic motion itself is slower than the purely electronic processes that underlie filamentary or charge-trapping switching, and that side reactions involving the mobile ions and the electrodes must be controlled by chemistry and by choice of contact metal. These costs are not prohibitive; they are the engineering problems of the route.

A fifth family, included here for completeness, is filamentary conduction in which metallic species — typically silver or copper cations electrochemically supplied by an active top electrode — migrate through the organic matrix and form conductive bridges analogous to those of inorganic conductive-bridge memories. Such filamentary organic devices have been demonstrated, and they inherit both the strengths (large on/off ratios, low switching voltages) and the weaknesses (stochastic filament formation, intra-device instability) of their inorganic counterparts discussed in section 1.4.1. They illustrate the point that the distinction between organic and inorganic memristors is not primarily about mechanism type: the same filamentary mechanism can be realised on either side of the organic–inorganic boundary, and the practical differences that follow are dominated by substrate and process compatibility rather than by a fundamental distinction in the switching physics. The mechanism classes that are genuinely distinctive of the organic landscape — ionic redistribution in polymer electrolytes, redox switching of macromolecular species, conformational switching of designed molecular units — are those in which the chemical specificity of the organic components actively contributes to the internal-state physics.

The distinction that matters for the argument of this thesis is therefore not the label *organic* versus *inorganic*, but what kinds of internal-state physics a given materials family actually makes accessible. Among the organic options, ion-mediated mixed ionic–electronic systems are particularly attractive because they couple a transparent internal state variable to a kinetics that is directly engineerable through chemistry. Within that subset, polymeric ionic hosts — polymer electrolytes — offer an additional advantage: the ionic transport kinetics are controlled by the host polymer segmental motion, which is itself a chemical design variable through the choice of backbone and the glass-transition temperature of the material. Polymer-electrolyte composites, in which a conjugated semiconducting polymer is blended with an ion-conducting polyether host loaded with a dissociated salt, therefore constitute a particularly promising subset of the broader organic memristor landscape. Their detailed treatment is the subject of section 1.5's successor section and is not developed further here. The present section limits itself to establishing that this subset can be identified on first-principles grounds, before any specific thesis-level chemistry is discussed.

### 1.5.3 Historical Development of Organic Memristors

The organic memristor literature as it developed over the last three decades is not the history of a single device concept refined through incremental improvement, but the history of several heterogeneous streams of work, each rooted in a different mechanism family and each pursued with a different architectural preference. A concise survey of this history is useful here not to enumerate papers but to identify, at the level of the field, which problems were progressively solved and which combinations of features remained unresolved at the point where the present thesis was initiated. The aim is to make the argument of the next section — that polymer electrolyte composites constitute a natural organising platform within the broader organic landscape — follow as a field-level observation rather than as a thesis-internal assertion.

An early and influential strand of work took the organic electrochemical cell as a starting point and asked whether the ionic-redistribution processes that give light-emitting electrochemical cells their slow turn-on and their history-dependent behaviour could be repurposed as a memristive functionality. Zakhidov and co-workers, working in the Malliaras group, reported a *light-emitting memristor* [55] in which an ionic $[Ru(bpy)_3]^{2+}$ complex provided both the emissive species and the mobile ionic component, and showed that the coupled optical and electrical response of the cell exhibited a genuine history dependence consistent with memristive operation. The significance of the demonstration, viewed from the present vantage point, was less that it produced a practical memory element than that it identified ionic redistribution in a polymer host as a legitimate and mechanistically interpretable route to memristive behaviour. The limitations of the specific implementation — limited reversibility, a small number of distinguishable states, and a device architecture optimised for emission rather than for precise conductance control — were characteristic of the early phase of the field and were addressed, in different ways, by the subsequent lines of work.

A parallel strand of work took the simple polymer thin film itself as the active element and asked what memristive behaviour, if any, could be obtained from modest compositions under standard two-terminal contacts. Lei and co-workers, for example, reported resistive switching in poly(vinyl alcohol) thin films [63] under flexible-substrate conditions. Devices of this kind were useful for establishing that memristive phenomena could be observed in inexpensive and straightforwardly processed polymer matrices, and contributed to the case that organic-electronic fabrication was compatible with resistive-switching functionality, but they were typically binary, often poorly interpretable at the mechanistic level, and rarely offered the kind of analogue multi-state operation that is relevant for synaptic applications. They are cited here as part of the broader pattern rather than as foundational contributions to the ion-mediated route developed below.

A distinct and very productive strand of work adopted the organic electrochemical transistor as the architectural platform for synaptic emulation. This was the three-terminal side of the field, and it benefited from the considerable prior investment of the bioelectronics community in OECT design and modelling. Gkoupidenis and colleagues demonstrated that PEDOT:PSS OECTs with a liquid electrolyte gate could reproduce short-term plasticity, paired-pulse facilitation, and long-term potentiation and depression in response to appropriate gate-pulse protocols [64], and later extended this to devices with global electrolyte connectivity approximating the behaviour of a network of coupled synapses [65]. Xu and colleagues produced core–sheath nanowire synaptic transistors with controlled plasticity [66], and van de Burgt, Fuller, and their co-workers developed the non-volatile organic electrochemical neuromorphic device (ENODe) [12, 13], in which a dedicated non-volatile gate reservoir supplied a reproducible, programmable, and linearly updateable synaptic weight. These devices demonstrated very convincingly that organic electrochemistry could support the whole range of synaptic learning rules, with low operating voltages, good analogue control, and in some cases excellent linearity and symmetry. They also clarified, at a mechanistic level, the central role of ionic exchange between a polymer electrolyte and a conducting polymer channel as the physical origin of the synaptic behaviour [14]. Complementary demonstrations of OECT-based synaptic devices with halide-perovskite gates [67] extended the range of accessible timescales and underscored that the three-terminal architecture was flexible at the materials level.

From the perspective of the present thesis, the three-terminal line of work is of considerable intellectual importance but is architecturally distinct. An OECT synaptic device comprises a gated channel, a separate electrolyte, and in the non-volatile variant a dedicated programming reservoir; it is read through a drain current at a small bias while programming occurs through a gate at a different potential. This architecture permits excellent decoupling of read and write, which is why the ENODe family achieves its high linearity and symmetry, but it also imposes an overhead on each synaptic element — at minimum a third terminal and a contacted electrolyte volume — that is not trivial to scale to the densities of a passive crossbar. For many applications, and in particular for the heterogeneous reservoir and coincidence-detection paradigms developed in section 1.3.5, a simpler two-terminal element whose ionic physics is contained within the active layer itself is an attractive complement.

The two-terminal line of organic memristor work proceeded in parallel. A representative demonstration from this line is the biomimicking redox organic memristor reported by Liu and co-workers [52], in which an organic redox system embedded in a polymer matrix produced, in a two-terminal device, a short-term-to-long-term memory transition of the kind observed in biological synapses under repeated stimulation. The device exemplified two features of interest: it produced a continuous, history-dependent conductance evolution in a genuinely passive architecture, and it exhibited coexistence of short-term and long-term memory regimes as a function of stimulation strength. It also illustrated, through its own limitations, a characteristic difficulty of the purely redox-driven two-terminal route: redox processes are not always cleanly reversible, and the long-term regime in such devices sometimes reflects partially irreversible redox transformations rather than controlled state evolution. The question of how to obtain analogue, reversible, multi-state behaviour in a two-terminal organic device without relying on permanent redox changes was, at the time, a live one.

Two further lines of organic synaptic work, closely related in time, should be noted in this survey. The first is the development of stretchable and mechanically compliant organic synaptic devices. Kim and co-workers reported a bioinspired flexible organic artificial afferent nerve [50] that integrated pressure sensing and synaptic transmission in a mechanically compliant form factor, and Park and co-workers surveyed the broader landscape of flexible and biointerfaced synaptic devices in their review on *nervetronic* applications [51]. These contributions are important because they demonstrated that the mechanical and substrate-level advantages of organic processing are in fact realisable at the device level, and because they framed a class of applications — conformable, skin-mounted, or biointerfaced synaptic hardware — for which soft-matter devices are a natural rather than a contingent choice. The second is the development of molecular memristors with engineered redox and conformational behaviour. Goswami and co-workers demonstrated that a carefully designed metal-coordinated azo-aromatic molecular layer could support resistive memory with high on/off contrast and robust endurance [53], and subsequently extended this to molecular-scale memristive devices with a level of state control approaching that of genuine analogue multi-state operation [54]. These contributions are important because they establish the upper bound of what molecular-level design can achieve in an organic memristor, although the demanding synthetic chemistry involved places these devices in a different practical regime from the composite devices considered below.

The field also accommodates hybrid systems in which organic and inorganic components coexist in the active layer. Perovskite–organic hybrid memristors and synaptic devices [68], for example, exploit the efficient ionic conduction of halide perovskites together with organic charge-transport layers, and illustrate that the organic and inorganic routes are not mutually exclusive at the device level. Conjugated polymer systems with non-volatile resistive-switching character have also been explored [69], and an influential strand of work at the interface of organic electrochemistry and reservoir computing [26] has demonstrated that mixed ionic–electronic organic devices can serve as the active elements of a physical reservoir, in the sense of section 1.3.5, without requiring their volatility to be eliminated. These contributions confirm, from outside the narrow scope of any single architecture, that the ion-mediated route within the organic landscape is intrinsically well matched to fading-memory computation.

What the historical record shows, taken as a whole, is a field that progressively solved several of the individually challenging problems associated with organic memristive operation. Ion-mediated mechanisms were identified and interpreted in both three-terminal and two-terminal architectures. Analogue, linear, symmetric programming was demonstrated in the three-terminal, electrochemically gated family. Short-term-to-long-term memory transitions were demonstrated in two-terminal redox devices. Mechanical compliance was demonstrated in stretchable synaptic devices, and molecular-level control was demonstrated in engineered molecular memristors. Field-level reviews by the early 2020s [14] captured this picture and articulated organic neuromorphic electronics as a recognised subfield with several active design lines.

What the historical record does *not* show, by the same point in time, is a mature consolidation of these individually demonstrated features into a single two-terminal platform. Reversibility, multi-state analogue behaviour, two-terminal architectural simplicity, and a clearly interpretable ionic-physics mechanism had each been demonstrated, but not in combination within a common materials family with a transparent set of chemical design knobs. The three-terminal electrochemical family achieved reversibility and analogue control at the cost of architectural complexity, the two-terminal redox family achieved architectural simplicity but often at the cost of full reversibility, the molecular family achieved precision at the cost of synthetic tractability, and the flexible–stretchable family demonstrated mechanical compatibility without committing to a particular mechanistic route. Importantly, the thesis does not claim that the present work closes all of these gaps in one step. It claims that the specific gap in which a two-terminal, reversible, multi-state, ion-mediated device with interpretable chemistry could be built from standard organic-electronic components was still open at the point where this work began, and that the route most likely to close it was a composite of a conjugated semiconducting polymer with an ion-conducting polymer host and a dissociated alkali-metal salt.

The natural question at field level is therefore the one posed at the end of section 1.5.2: within the heterogeneous organic memristor landscape, which materials architecture provides the cleanest coupling between controllable ionic dynamics, reversible electronic response, and two-terminal passive operation? The answer defended in the remainder of this chapter is that polymer electrolyte composites, and specifically conjugated-polymer / polyether / alkali-salt composites processed from a common solution, constitute such an architecture. The next section develops that case in detail, first by reviewing the principles of ion-conducting polymers, then by introducing light-emitting electrochemical cells as the well-characterised precedent for ionic redistribution in a conjugated-polymer host, and finally by articulating the rationale for adopting the composite approach as the enabling strategy of the thesis.

## 1.6 Polymer Electrolyte Composites: The Enabling Strategy

The argument of section 1.5 narrowed the organic memristor landscape to the subset of devices in which ionic redistribution in a polymer host couples to electronic transport in a conjugated semiconductor. Within that subset, the remaining design decision is architectural: how the ionic and electronic functions are physically organised inside the active layer. Three-terminal electrochemical transistors resolve this by placing the ion reservoir in a separate, contacted electrolyte volume and by gating the channel through an independent electrode. Two-terminal devices must integrate both functions into a single stack driven by a single bias. The enabling strategy adopted in this thesis, and the one pursued in the remainder of this chapter, is to build that single stack as a *composite*: a semiconducting conjugated polymer blended with an ion-conducting polyether host and a dissociated alkali-metal salt, processed from a common solution onto an ordinary two-terminal electrode stack.

The case for this strategy rests on three arguments, developed in turn in the following subsections. section 1.6.1 recalls the physical-chemistry principles of ion-conducting polymers, the role of polyether hosts in coordinating alkali cations, the main transport mechanisms in such hosts, and the compositional variables that set the ionic response. section 1.6.2 introduces polymer light-emitting electrochemical cells as the established precedent for stable, reversible, field-driven coupling between ionic migration and electronic transport in a soft organic active layer, and identifies the specific subset of this precedent that is directly material-level relevant to the thesis. section 1.6.3

then articulates the rationale for the composite approach itself: the decoupling of electronic and ionic functions, the choice of salt as an independent design variable, the treatment of cation identity and composition as chemically controlled timescale knobs, and the implication that a small family of chemically distinct composites can constitute a temporal-kernel bank rather than a set of competing candidate memories.

### 1.6.1 Ion-Conducting Polymers: Principles

An ion-conducting polymer, in the sense relevant to this thesis, is a macromolecular host in which dissolved or dispersed ionic species are mobile under an applied electric field without requiring a liquid solvent phase. The field was opened by the recognition, in the late 1970s and early 1980s, that alkali-metal salts could be dissolved in poly(ethylene oxide) (PEO) to form solid complexes that conduct ions at macroscopically measurable rates [70–72]. The subsequent development of solid polymer electrolytes as a field, motivated primarily by lithium battery applications, produced a consolidated set of principles that this subsection summarises at the level appropriate for an introductory chapter. The discussion is deliberately restricted to the features that make polymer electrolytes a tunable memristive platform; no attempt is made to review the full solid-polymer-electrolyte literature.

The central chemical feature of a polyether host is the presence of a high density of ether oxygens along the polymer backbone or its side chains. In PEO itself, the oxygen atoms are spaced at regular intervals along a $-CH_2-CH_2-O-$ repeat unit, producing a linear backbone whose conformation in the amorphous phase allows each cation to coordinate simultaneously to several ether oxygens drawn from one or more polymer chains [71, 73]. In hyperbranched polyester-amide hosts of the Hybrane family, the oxygen density is provided instead by a combination of ether and ester side-chain groups grafted onto a branched polymer core. In both classes of host, the relevant chemical role of the oxygen atoms is the same: they act as hard Lewis bases that solvate alkali cations, stabilise the dissociated state of the salt, and determine the local binding environment in which the cation resides. The anion is held more weakly, primarily by electrostatic compensation of the cation, and typically exhibits higher mobility than the cation within the same host, especially when it is a soft, charge-delocalised species such as triflate ($CF_3SO_3^-$) or bis(trifluoromethanesulfonyl)imide (TFSI, $N(SO_2CF_3)_2^-$).

Ion transport in such hosts is most naturally described by a combination of three microscopically distinct mechanisms. In the *vehicular* picture, a cation together with its primary solvation shell of coordinated oxygens moves through the polymer matrix as an effective ionic vehicle; long-range transport requires the coordinated oxygens themselves to move, which couples ionic transport to polymer motion. In the *hopping* or site-to-site picture, the cation is transferred between nominally equivalent coordination sites on the host by a local rearrangement of the oxygens that does not require macroscopic transport of the host itself; this process is closely related to the Grotthuss-type description familiar from protonic conductors, although the chemistry of the transfer event is different. In the *segmental-motion-assisted* picture, the reorganisation of short polymer segments around the cation — segmental motion of the backbone or of the side chain — generates and destroys coordination sites near the ion in such a way that the effective diffusion of the ion is enabled by the local dynamics of the host. These pictures are not mutually exclusive; in a realistic polyether host, transport is a hybrid of all three, and their relative weights depend on the host chemistry, on the cation, on the salt concentration, and on temperature [71, 72]. What matters for the present thesis is the qualitative implication that ionic transport in a polyether host is not set by the polymer chemistry alone, but by the coupling between the polymer dynamics and the ionic chemistry.

A consequence that is essential for the remainder of the thesis follows directly from this coupling. The macroscopic ionic mobility in a polymer electrolyte is, to a first approximation, controlled by the segmental mobility of the host, which is in turn controlled by the glass transition temperature $T_\mathrm{g}$ and by the degree to which the host remains amorphous under operating conditions. Below $T_\mathrm{g}$, segmental motion is effectively frozen and ionic conduction collapses; above $T_\mathrm{g}$, segmental motion is activated and ionic transport becomes measurable, with a characteristic Vogel–Tamman–Fulcher rather than Arrhenius temperature dependence in many systems [72]. Crystalline regions, where they exist, contribute little to conduction and may in fact trap ions in ordered environments; for this reason, low-crystallinity or fully amorphous hosts are preferred for ion-transport applications. Engineering the host for a low $T_\mathrm{g}$ and a low crystalline fraction is a recognised design strategy in solid polymer electrolytes, pursued through copolymerisation, branching, and plasticisation, and hyperbranched architectures such as the Hybrane family are explicitly motivated in part by the suppression of crystallinity that branching entails. For memristive operation at room temperature, the practical implication is that the host should remain amorphous and segmentally mobile under normal ambient conditions, and that these properties are accessible through synthetic design at the host level without recourse to external plasticisers.

The strength of the interaction between a cation and the oxygen donor atoms of the host is determined by the chemical identity of the cation, and it is here that hard–soft acid–base (HSAB) reasoning [74] becomes a useful intermediate-level framing. The ether and ester oxygens of polyether and polyester-amide hosts are hard Lewis bases: their donor electron pairs are localised, their polarisability is low, and the coordination bonds they form with cations are correspondingly dominated by electrostatic rather than covalent contributions. Among alkali cations, $Li^+$ is the hardest Lewis acid — small radius, high charge density, highly localised charge — and therefore interacts most strongly with a hard-base oxygen donor. $Na^+$ and $K^+$ are progressively softer in this sequence, with larger radii, lower charge densities, and weaker electrostatic coupling to a given oxygen environment. The practical consequence is that, within a common polyether host at a fixed composition, the characteristic residence time of the cation at a given coordination site, and therefore the kinetics of its displacement under an applied bias, are expected to follow the ordering $Li^+ > Na^+ > K^+$ in terms of coordination strength. In dynamical terms, this ordering is expected to map onto a corresponding ordering of the fading-memory timescale of the device, with harder cations producing slower relaxation and softer cations producing faster relaxation. The HSAB framing is used here as a qualitative organising principle — it anticipates a ranking of timescales without fixing quantitative values — and the quantitative confirmation of this ordering against real measurements is explicitly reserved for the experimental chapters.

Composition provides the second independent family of design knobs. A polymer electrolyte is characterised not only by its host chemistry and its cation identity, but also by the ratio of salt to host and, when it is embedded in a larger composite, by the ratio of polymer electrolyte to semiconducting phase. These ratios control, respectively, the concentration of mobile ionic species available for redistribution under a bias, and the effective volume fraction of the electronic and ionic phases within the active layer. The two ratios are not redundant: the salt-to-host ratio sets the number and the spatial density of cation coordination sites that are occupied, while the host-to-semiconductor ratio sets the volume available for ionic transport and for electronic percolation through the conjugated phase. In the composite devices considered in later chapters of this thesis, the polymer-to-semiconductor mass fraction and the salt-to-semiconductor mass fraction will therefore be treated as two independent axes of the composition space, and their effects on the dynamical response of the device will be distinguished experimentally. At this stage, the essential point is that composition is a real physical-chemistry knob on the ionic dynamics, not a parameter selected for convenience.

The combination of these considerations motivates the specific host choices made in the thesis. Hybrane DEO750 8500, a hyperbranched polyester-amide with a high density of ether and ester oxygens on its side chains, offers an amorphous, low-crystallinity host that has already been validated in the same material family used in **??** and in its associated light-emitting precedent [62]; it is used as the ion-transport host in the proof-of-concept SY/Hybrane/LiTf composite, where it is paired with a triflate salt to demonstrate the full set of memristive functionalities in a single device. PEO is introduced as the central polyether host for the broader comparative corpus explored later in the thesis, where it plays a chemically analogous role: its ether oxygens coordinate alkali-metal cations, its segmental motion enables ionic transport, and its compositional and cation-identity dependencies will be varied systematically. The intention of the thesis is not to argue that one of these hosts is superior to the other, but to use them for complementary purposes: Hybrane as the host in which the full synaptic functionality is established on a single exemplar, and PEO as the host in which composition- and cation-driven dynamics are explored across a broader comparative set. The shared physical-chemistry principles developed in this subsection apply to both.

### 1.6.2 Light-Emitting Electrochemical Cells as Precedent

The argument developed so far relies on a non-trivial physical claim: that ionic redistribution in an organic polymer host can be coupled to electronic transport in a conjugated semiconductor in a manner that is reversible, stable over many operating cycles, and reproducible at the device level. This claim is not hypothetical. It is directly supported by several decades of work on polymer light-emitting electrochemical cells (LECs), a device family in which exactly this coupling is exploited for a different practical purpose — electroluminescence — and in which the relevant physics has been characterised in considerable detail. LECs are invoked here not because the thesis is concerned with light emission, but because they constitute the cleanest experimental demonstration that mixed ionic–electronic composites of the kind envisaged in the previous subsection are a reproducible and stable device class. Their relevance is as a *material-level precedent*; their relevance as a device class is not pursued.

An LEC in its canonical form [56] is a two-terminal sandwich device in which a conjugated light-emitting polymer is blended, from a single solution, with an ion-conducting polymer and a dissociated salt. Under an applied bias, ions in the host redistribute: cations drift toward the cathode and anions toward the anode, producing regions of electrochemically doped polymer in the neighbourhood of each contact. The doped regions supply, respectively, electrons and holes to the undoped bulk, and the resulting p-i-n-like profile develops dynamically in response to the applied bias. The intrinsic region between the two doped fronts is where electron–hole recombination occurs and where electroluminescence is generated. Microscopically, the operation of an LEC has been the subject of long-running debate between an electrochemical-doping picture, in which the doping of the polymer is the dominant effect, and an electrodynamic picture, in which the accumulation of ions at the interfaces acts primarily to reduce the charge-injection barriers. Detailed modelling [57, 58] and experimental studies [59, 61, 75] clarified the conditions under which each picture dominates, and a unifying framework [60] subsequently showed that the two views are not contradictory but correspond to different injection regimes: the electrodynamic picture applies when charge injection is barrier-limited and ionic accumulation restructures the interfaces, while the electrochemical-doping picture applies when injection is ohmic and the doped regions themselves dominate the transport. Both mechanisms involve the same underlying ionic redistribution in the polymer host; they differ in which consequence of that redistribution controls the measurable electrical behaviour. Complementary studies of electrochemical side reactions [76] have identified the chemical routes by which LECs degrade and have informed strategies to suppress them.

Two aspects of this precedent are directly relevant to the thesis. The first is that LEC operation demonstrates, across several orders of magnitude of applied bias and over operational lifetimes that can reach hundreds or thousands of hours, that ionic redistribution in a polymer host coexists stably with electronic transport in a conjugated polymer phase. The ions do not, in general, undergo irreversible side reactions that destroy the device within a few cycles; the conjugated polymer does not, in general, suffer irreversible electrochemical degradation under the normal operating window; and the interfaces between the polymer composite and the metallic contacts can be engineered, through choice of metal and of operating protocol, to support repeated injection and withdrawal of both charge and coordinated ions. These are empirical findings, established in the LEC literature, and they establish the upper-level feasibility of mixed ionic–electronic composite operation as a stable device mode. For a memristive device based on the same composite design, these findings translate directly into a plausibility argument: if ionic redistribution in a polymer host is stable under LEC operating conditions, then its use as the internal state variable of a two-terminal memristive device is not intrinsically precluded by chemistry.

The second relevant aspect is that the solid electrolyte in an LEC plays an enabling role that is conceptually analogous to the role of the ion-conducting phase in a memristive composite. In both cases, the electrolyte is not the electroluminescent or the electronically conducting component, and in both cases its contribution to the device response is mediated entirely through the redistribution of its mobile ions under the applied bias. The difference between the two applications is in how the redistribution is exploited: in the LEC, the driving bias is held during operation so that the doped p-i-n profile is maintained and emission is sustained; in the memristive device, the bias sets an internal ionic distribution whose residual after the bias is removed carries the memory of the stimulation. The chemistry of the electrolyte, its coupling to the conjugated host, and its stability under field-driven operation are, however, governed by the same principles in both cases. The LEC literature therefore offers a well-characterised, long-developed experimental testbed for the material chemistry that this thesis adopts for a different purpose.

The strongest form of this precedent, for the present work, is the specific case of LECs built on the same material family that is used for the proof-of-concept device in **??**. Mardegan and co-workers [62] reported that hyperbranched polyester-amide hosts of the Hybrane family, blended with a conjugated light-emitting polymer, support LEC operation with long operational lifetimes and well-controlled electroluminescence characteristics. This demonstration provides a materials-level validation for the same semiconducting-polymer / polyether / salt composite architecture that is later used in this thesis to produce a memristive device: the constituent polymers, the solvent system, the processing protocol, and the operating conditions are directly comparable. The thesis adopts this precedent specifically: the Chapter 2 device is built on the same Hybrane-class host demonstrated in the LEC work, and the general feasibility of reversible field-driven ionic redistribution in that host class is therefore not a conjecture but a transfer of established material knowledge into a new device function.

It is important to be clear about what this precedent does and does not imply. LEC operation establishes that stable coupling between ionic and electronic functions in an organic composite is achievable; it does not guarantee that the specific quantitative synaptic behaviour required of a memristive device — analogue multi-state programming, short- and long-term memory regimes, clean relaxation kinetics — follows automatically from any LEC-validated composite. The proof-of-concept chapter takes up precisely this question for the SY/Hybrane/LiTf system. The subsequent comparative chapters extend the inquiry to a PEO-based polyether host across a range of compositions and cations, and do not assume that every host–salt–cation combination that is chemically plausible will reproduce the full behaviour of the Chapter 2 exemplar. The LEC precedent sets the upper-level chemistry on a firm footing; it does not substitute for the detailed memristive characterisation pursued in later chapters.

### 1.6.3 Rationale for the Composite Approach

The two preceding subsections set out the physical-chemistry principles of ion-conducting polymers and the material-level precedent that such polymers can stably support field-driven ionic redistribution in a soft organic device. The central design decision of this thesis can now be stated directly: instead of seeking a single molecular or polymeric species in which electronic conduction, ionic transport, and controlled coupling between the two are all realised simultaneously, the thesis adopts a *composite* active layer in which each of these functions is supplied by a different component. A conjugated polymer provides the electronic transport pathway; a polyether host provides the ionic-transport matrix and the coordination environment for the mobile cation; and a dissociated alkali-metal salt provides the mobile ionic species itself. The three components are blended from a common solution and deposited as a single thin film between two electrodes. The memristive response of the device is then produced by the coupling between the electronic transport of the conjugated phase and the residual ionic distribution in the polyether phase, and its dynamics are set by the chemistry of each component and by the composition of the blend.

The principal advantage of this approach is that it decouples the engineering problem into three tractable chemistry subproblems. The electronic transport is the responsibility of the conjugated polymer, which can be chosen from the already-mature set of conjugated semiconductors developed for organic electronics, and whose transport properties are reasonably well understood. The ionic transport is the responsibility of the polyether host, which can be chosen from the established set of solid polymer electrolytes, and whose coordination chemistry and segmental dynamics are amenable to systematic investigation as outlined in section 1.6.1. The provision of mobile ions is the responsibility of the salt, which is chosen by selecting the cation and anion independently from their respective chemical series. The alternative — seeking a single monolithic material that supplies all three functions at once — is a more constrained design problem in which any adjustment to one property tends to couple to the others and in which the relevant variables are not, in general, independently tunable. The composite approach is chosen precisely because it restores that independence: each component can be varied with the others held approximately fixed, and the resulting changes in device dynamics can be attributed, with reasonable confidence, to the component that was changed.

The decoupling is particularly valuable in the case of the salt, which becomes, in this framing, an independent design variable whose relevance to the device dynamics is disproportionate to the fraction of the active layer that it occupies. In a composite with a common host polymer, the choice of cation sets the strength of the coordination environment that the mobile ion experiences, and therefore the kinetics of its redistribution under bias. Hard Lewis acid cations such as $Li^+$ are coordinated strongly to the ether oxygens of the host and are therefore difficult to displace and slow to relax; softer cations such as $Na^+$ and $K^+$ are coordinated less strongly and are therefore expected to respond on shorter timescales, as anticipated in section 1.6.1. The choice of anion enters differently. In a polyether host at a fixed cation, the anion is held primarily by electrostatic compensation of the cation and is typically a soft, charge-delocalised species; its mobility is usually higher than that of the cation and its residence time on any given host site is correspondingly shorter. The asymmetry between the mobility of the cation and that of the anion is therefore an intrinsic feature of the chemistry of the salt in a polyether host, and it is not an accidental property of any particular device.

This asymmetry is itself a design resource. If the cation relaxes slowly and the anion relaxes quickly, then the composite naturally carries two distinguishable memory timescales within the same active layer: a fast component associated primarily with anion redistribution at the interfaces, and a slow component associated primarily with cation redistribution in the bulk of the polyether host. The relative contribution of each component to the device response under a given stimulation protocol depends on the amplitude and duration of the applied bias and on the read protocol, and separating the two regimes experimentally is non-trivial, but the chemical origin of the two-timescale behaviour is transparent. It is this same chemical asymmetry that was used, in section 1.2.2 and in the broader synaptic-plasticity literature [30, 31, 33], to motivate the coexistence of short-term and long-term plasticity regimes in a single biological synapse. The memristive analogue of that coexistence, in the composite architecture, is a direct consequence of the salt chemistry and does not require any additional device structure.

The compositional variables of the composite extend the design space further. Within a given host–salt combination, the polymer-to-semiconductor and salt-to-semiconductor mass fractions independently control the ionic content and the ionic-transport volume, and therefore independently control the amplitude and the rate of the ionic redistribution under a given bias. A composite with a higher salt fraction provides more mobile ions and a stronger ionic response per unit volume; a composite with a higher polyether fraction provides a larger segmentally-mobile matrix in which ions can rearrange. These two knobs are not equivalent, and in principle they should separate the amplitude of the programmable conductance change from the kinetics with which that change develops and relaxes. The experimental exploration of this separation is deferred to the comparative chapter of the thesis; here, the point is only that composition is a real and independent variable of the composite architecture, not a nuisance parameter to be tuned to a single optimum.

One-step blend processing is, in this context, not merely a fabrication convenience but a feature that makes systematic compositional investigation possible. Because the semiconductor, the host polymer, and the salt are all soluble in a common organic solvent, a composite device of arbitrary composition can be produced by mixing the three stock solutions in the desired proportions and depositing the resulting blend by a single spin-coating or blade-coating step onto a patterned electrode. Changes in composition correspond to changes in the mixing ratio and do not require redesign of the device stack. The fabrication effort per composition is therefore modest, and the difference between nominally identical compositions is controlled by a small number of well-defined mixing and deposition parameters rather than by the reproducibility of a multi-step process. This is what makes it practical, for the comparative chapter of the thesis, to build a set of compositions and cations within a common host and to characterise them on a common experimental basis.

The implication of this analysis is that the main chemical variables of the composite — the identity of the mobile cation, the identity of the anion, and the polymer and salt mass fractions — are not simply materials parameters whose selection yields a more or less capable device. They are handles on the internal-state physics of the device, and specifically on the characteristic timescale of its fading memory. A choice of cation sets a default timescale through its coordination strength; a choice of composition scales the amplitude and the rate at which the ionic response develops; the resulting device, read with a standard electrical protocol, exhibits a dynamical response whose characteristic time reflects these choices. In this framing, cation identity and composition are *computational* design knobs as much as they are chemical ones: they select the fading-memory constant that the device presents to its circuit environment. The thesis adopts this framing as its central analytical move and pursues it, in the experimental chapters, by explicitly quantifying how composition and cation chemistry map onto measurable device dynamics.

A further consequence is that the composite approach supports a distinctive computational use pattern that would be awkward to obtain from a homogeneous material. Because the three chemical variables above can be varied independently and because the fabrication is a single-step blend deposition, a set of chemically distinct composites can be prepared, each with a different expected timescale, and assembled together as a small family of devices. The members of the family share the same electronic transport chemistry and the same host polymer but differ in composition or in cation identity; their dynamical responses, as anticipated by the physical chemistry above, are expected to span a range of fading-memory timescales. Such a family, when read out through a common circuit, constitutes in effect a bank of temporal kernels with different decay constants — precisely the substrate required for the heterogeneous physical reservoir computing, the coincidence-detection, and the multi-timescale transient filtering paradigms discussed in section 1.3.5. The heterogeneity that would be an obstacle for a conventional trained-weight application is, in this framing, the mechanism by which temporal-computing functionality is obtained in hardware. This observation will be developed quantitatively in the application-oriented part of the thesis; here it is stated only as the natural computational counterpart of the compositional freedom that the composite architecture provides.

The practical programme that follows from this design rationale can now be articulated at the level of the thesis. The proof-of-concept (**??**) demonstrates, on a single composite device built on a hyperbranched polyester-amide host with a lithium triflate salt, that the composite architecture supports a complete set of memristive functionalities and that the two-timescale behaviour anticipated from the cation–anion asymmetry is in fact observed. The comparative experimental chapter then takes the composite as fixed in its electronic component and its host chemistry, and sweeps the composition and the cation identity within a PEO-based polyether host across the alkali-metal triflate series, on a common set of dynamical measurements, to test whether composition and cation act as predicted on the device timescale. The application-oriented chapter takes the dynamical characterisation obtained in the comparative chapter as its input and uses the resulting family of timescale-distinct devices as the substrate for temporal-computing demonstrations. Each of these chapters exercises a specific part of the design rationale developed here; the rationale itself, at the level of principle, is the subject of the present section, and the concrete experimental and computational instantiations belong to the chapters that follow.

It is worth stating clearly, before closing, the boundary of what the composite rationale does and does not claim. It does not claim that any particular composition will produce a superior non-volatile memory; the retention-oriented benchmarks of non-volatile storage are not the target of the thesis. It does not claim that every host–salt–cation combination will produce the same synaptic richness as the proof-of-concept device; the full synaptic suite is an exemplar result and is not propagated wholesale onto the comparative corpus. It does not claim that the thesis establishes a complete physical-chemistry theory of polymer-composite memristive operation; what it establishes, instead, is a programmatic link between chemical design variables and dynamical device behaviour, supported quantitatively in the specific regions of the design space where experimental evidence is abundant. The claim of the rationale is therefore a positive and specific one: polymer electrolyte composites provide an enabling architecture for coupling ionic dynamics to electronic transport, their dynamics can be engineered through host chemistry, salt identity, and composition, and the thesis is now in a position to state its objectives as a concrete programme built on that enabling strategy.

## 1.7 Scope and Objectives of This Thesis

The preceding sections built a chain of arguments from the limits of conventional computing through the physiology of biological synapses, the memristor as the relevant hardware abstraction, the inorganic and organic memristor landscapes, and the polymer-electrolyte composite as an enabling architecture for chemically tunable ion-mediated dynamics. Each of those steps was developed at field level. This closing section converts that field-level argument into the specific scientific questions that the thesis addresses, the specific objectives it pursues, and the structure through which those objectives are realised across the remaining chapters.

### 1.7.1 Open Scientific Questions

The arguments developed in sections 1.5 and 1.6 converge on a small number of scientific questions that organise the work of the thesis. They are stated here in the form actually addressed in the chapters that follow rather than as a generic catalogue of open problems.

The first question is whether a fully reversible two-terminal organic memristive device can be realised from a solution-processed polymer-electrolyte composite using only standard organic-electronic fabrication. The historical survey of section 1.5.3 identified this combination — two-terminal architecture, full reversibility, multi-state analogue control, and a chemically interpretable ionic mechanism — as a gap in the organic memristor landscape at the start of the present work. The second question, contingent on the first, is whether such a device can support a substantive range of synaptic functionality in a single platform: analogue switching, distinguishable short- and long-term memory regimes, an excitatory-post-synaptic-current response, spike-timing-dependent plasticity, and an impedance signature consistent with the proposed mechanism. The third question concerns the physicochemical origin of the memory regimes themselves: whether the coexistence of a fast and a slow memory component in the composite can be attributed to the cation–anion mobility asymmetry anticipated in section 1.6.3, and whether the chemistry of the cation–host coordination accounts for the slow component on a hard-soft acid-base basis.

The next set of questions broadens the inquiry from a single exemplar to a comparative experimental basis. Within a common polyether host — specifically the linear polyether PEO — and a common alkali-metal triflate salt family, how does composition affect the dynamical response of the device when the polymer-to-semiconductor and salt-to-semiconductor mass fractions are varied around a reference formulation? Holding the composition fixed instead at a reference value, how does substitution of the cation along the $Li^+$, $Na^+$, $K^+$ series shift the fading-memory timescale measured by a common dynamical protocol? These questions are pursued through three measurements that are abundant across the comparative corpus — I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation — and the answers reported in the thesis are anchored exclusively in those three measurements. The richer synaptic suite (EPSC, STDP, separated short- and long-term memory retention, impedance) is treated as a Chapter 2 result and is not propagated as a uniform comparative claim.

The final set of questions belongs to the application axis of the thesis. Can volatile organic memristors of the kind developed here be exploited as temporal-computing primitives in their own right, rather than judged against the figures of merit appropriate to non-volatile memory? Can composition and cation identity be treated as design knobs for application-level timescale matching, in the sense that a deliberately heterogeneous family of devices spans a controlled range of fading-memory time constants? Can the device-to-device and cycle-to-cycle variability of such a family, characterised quantitatively from the same common measurements, contribute usefully to the dynamical richness of a temporal-computing scheme rather than be treated only as a defect to be suppressed? These questions are answered in simulation in the data-driven application chapter, using behavioural models extracted from the comparative experimental corpus and using the proof-of-concept device only as a source of priors and sanity checks.

### 1.7.2 Thesis Objectives

The scientific questions of section 1.7.1 translate into the following ordered objectives, which structure the remaining chapters of the thesis. They are stated explicitly to make the scope of the work, and the boundary of what it does not claim, transparent from the outset.

1. Design, fabricate, and characterise a proof-of-concept two-terminal organic memristive device based on a SY/Hybrane/LiTf composite, processed from a single solution onto a standard ITO/composite/Ag stack, and demonstrate the basic memristive functionality in a single platform.

2. On that single exemplar device, demonstrate the full set of synaptic functionalities relevant to the thesis: analogue multi-state switching, distinguishable short-term and long-term memory regimes with separately measured retention curves, excitatory-post-synaptic-current behaviour under pulsed stimulation, spike-timing-dependent plasticity with a biologically plausible time constant, and an impedance signature consistent with the proposed mechanism.

3. Provide a mechanistic account of the two memory regimes of that exemplar in terms of the displacement of mobile triflate anions and lithium cations within the polyether host, supported by hard-soft acid-base reasoning on the cation–oxygen interaction.

4. Build a comparative experimental corpus on a PEO-based polymer-electrolyte platform loaded with alkali-metal triflate salts, fabricated under a common protocol, and characterise it through three common dynamical measurements — I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation — that are abundant across the corpus.

5. Within that corpus, quantify (a) the composition dependence of the dynamics through a PEO/LiTr concentration series in which the polymer and salt mass fractions are varied around a reference composition, and (b) the cation dependence through a PEO/LiTr, PEO/NaTr, PEO/KTr set at a fixed reference composition, on the same three common measurements.

6. Treat TMPE-based composites and alkali-bis(trifluoromethanesulfonyl)imide ($LiN(SO_2CF_3)_2$, $NaN(SO_2CF_3)_2$, $KN(SO_2CF_3)_2$) systems transparently as exploratory side evidence with a clearly delimited dataset, used to motivate or to cross-check specific trends, and not as parallel main platforms.

7. Construct compact, experimentally grounded behavioural models per composition and per cation from the three common dynamical measurements of the comparative corpus, with the proof-of-concept device used only as a source of priors and sanity checks where its richer synaptic data are directly relevant; the models capture a fading-memory time constant, a nonlinear pulse-update rule, a device-to-device variability envelope, and a read transfer function.

8. Demonstrate, in simulation and from those models alone, three application-level temporal-computing schemes for which the composition- and cation-driven timescale spread of the comparative corpus is a natural substrate: a heterogeneous physical reservoir for temporal classification, a spike coincidence detector built on the measured variable-delay depotentiation kernels, and a multi-timescale transient filter bank built on devices with deliberately distinct fitted time constants.

These objectives reflect the actual evidence base of the thesis. Objectives 1–3 are exemplar-level and rely on the full synaptic measurement suite of the proof-of-concept device. Objectives 4–6 are comparative and rely strictly on the three common dynamical measurements; they do not assume that the richer synaptic metrics are available across the comparative corpus. Objectives 7–8 are computational and are framed as data-driven exploitation of measurements already acquired in the experimental chapters, rather than as detached theoretical work; in particular, no further fabrication is required to fulfil them.

### 1.7.3 Thesis Outline

The thesis is organised in five chapters that follow the logical progression of the objective list. Each chapter is described below at the level of structure and scientific contribution; its specific contents are developed in the chapter itself.

**Chapter 1 — Introduction.** The present chapter has built the conceptual and materials background required to motivate the rest of the thesis. It reviewed the limits of conventional computing architectures, identified the biological synapse as the engineering template for the relevant hardware, introduced the memristor as the appropriate device abstraction, surveyed the inorganic and organic memristor landscapes with explicit attention to the role of volatile fading-memory dynamics as a first-class computational element, and identified polymer electrolyte composites as the enabling material strategy for chemically tunable ion-mediated operation. The closing section, of which the present subsection is the final part, converts this material-level argument into the scientific questions and objectives pursued in the remainder of the thesis.

**Chapter 2 — Proof of Concept: Polymer-Based Composite Organic Memristive Devices.** The second chapter presents the experimental proof of concept of the thesis on a SY/Hybrane/LiTf composite device, fabricated from a single solution onto an ITO/composite/Ag stack, and demonstrates the full set of memristive and synaptic functionalities in that single platform. This is the only device of the thesis on which the complete characterisation suite — I–V hysteresis, potentiation and depression curves, EPSC, separated short- and long-term memory retention, spike-timing-dependent plasticity, and impedance spectroscopy — is treated as primary evidence. The chapter develops the mechanistic interpretation of the two memory regimes in terms of mobile triflate anions and lithium cations within the polyether host, and supports it with the hard-soft acid-base reasoning introduced in section 1.6.1. The chapter is built on the work originally reported in Prado-Socorro et al. [77] and is the richest synaptic dataset of the thesis.

**Chapter 3 — PEO–Triflate Polymer-Electrolyte Memristive Devices: Composition and Ion-Identity Dependence.** The third chapter takes the validated proof-of-concept architecture forward and reformulates it on a PEO-based polyether host loaded with alkali-metal triflate salts, in order to test the central design rationale of section 1.6.3 comparatively rather than on a single exemplar. The chapter is built around a core PEO/LiTr concentration series, in which the polymer and salt mass fractions are varied around a reference composition, and a secondary PEO/LiTr, PEO/NaTr, PEO/KTr set held at the reference composition for cation comparison. Both studies share three common dynamical measurements — I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation — and these three measurements define the comparative basis of the chapter and of the application-oriented chapter that follows. A short, clearly delimited section reports a small set of TMPE-based and alkali-bis(trifluoromethanesulfonyl)imide composites as exploratory side evidence and is presented as such. Variability is quantified directly from replicates within the three common datasets and is treated as a property of the device family rather than as a defect to be suppressed.

**Chapter 4 — Data-Driven Temporal Computing with Volatile Polymer-Electrolyte Memristors.** The fourth chapter converts the experimental dynamical measurements of the previous two chapters into application-level functionality, without further fabrication. Compact behavioural models are extracted, per composition and per cation, from the three common dynamical measurements of the comparative experimental chapter, and the proof-of-concept device of Chapter 2 is used only as a source of priors and sanity checks where its richer synaptic data are directly relevant. The chapter then demonstrates, in simulation, three application-level schemes in which the composition- and cation-driven timescale spread of the polymer-electrolyte family is the active resource: a heterogeneous physical reservoir for temporal classification, a spike coincidence detector grounded directly in the measured variable-delay depotentiation kernels, and a multi-timescale transient filter bank built on devices with deliberately distinct fitted time constants. Circuit-level integration constraints — 1T1M addressing, current compliance, read and write protocols, variability envelopes — are treated explicitly. The chapter is grounded in measured experimental data throughout; it is not a detached theory chapter.

**Chapter 5 — Conclusions and Outlook.** The closing chapter synthesises the contributions of the thesis along the device-chemistry axis (**??** and the comparative-experimental chapter) and the application axis (the temporal-computing chapter), benchmarks the work against other temporal, reservoir, and event-driven neuromorphic hardware rather than against digital memory technologies, states the limitations of the current evidence base honestly — in particular the restriction of the richer synaptic metrics to the proof-of-concept device and the exploratory status of the TMPE and alkali-bis(trifluoromethanesulfonyl)imide systems — and identifies the future directions that the present work most directly enables.

The narrative arc of the thesis can therefore be summarised as motivation in the present chapter, proof of concept in **??**, systematic comparative study in the chapter that follows, computational reinterpretation and exploitation in the chapter after that, and synthesis in the closing chapter. The thesis does not pursue, and does not claim to pursue, the construction of a competitive non-volatile memory; it pursues the construction of a chemically transparent route from polymer-electrolyte composition and cation chemistry, through measured ion-mediated device dynamics, to functional temporal-computing primitives. The chemistry-to-dynamics-to-computation chain is the central analytical structure of the work, and the chapters that follow develop it in turn.

The next chapter takes up the first link in that chain. It presents the proof-of-concept SY/Hybrane/LiTf device in detail, establishes that the composite architecture supports a complete set of memristive and synaptic functionalities in a single two-terminal platform, and develops the mechanistic interpretation that anchors all subsequent comparative and computational work.